Which 16 of 24? Per-Die Silicon Harvest Maps, Hardware Lots and Core Guarding from Supercomputer Telemetry

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***Abstract*—Vendors sell large dies as lower-core-count parts by disabling cores, defective or not, without saying which. On CINECA's Marconi100 the baseboard management controller exposes one thermal sensor per core position without renumbering survivors, so the positions that never report form each die's *harvest map*. From public telemetry we recover maps for 1,962 POWER9 sockets over 1.67 million socket-days. In every map the eight disabled cores of the 24-core die form four complete *slices*, core pairs sharing an L2/L3 block, though the hardware can disable single cores; 443 of 495 possible patterns occur, varying from die to die. Disabled slices cluster above all within the six *quads*, slice pairs adjacent in IBM's chip diagram, where thermal coupling is also strongest. A sharp boundary at rack 22, predictable out of sample, separates two populations of parts with different disable statistics, most plausibly procurement lots. An independent boot signal places a boot inside 133 of 136 steady-state map changes, and none without one. Nine steady-state core-guard episodes, removing a slice or core at boot, are each matched by the operating system's own CPU count. Maps travel with their dies, so map changes trace processors between nodes: sockets change die at 2.1 % per year, new processors entering at an inferred 1.0–1.2 %. Equivalence tests bound the map's association with socket power to a few watts, and it does not usefully predict replacement. The disclosure lies in which sensors exist, not their values, so coarser sampling does not remove it.**

***Index Terms*—core harvesting, silicon binning, HPC telemetry, POWER9, yield, hardware provenance**

I. Introduction

Large processor dies are susceptible to manufacturing defects [1], and a die whose faulty core is disabled can still be sold [2], [3]; vendors also disable working units to fill cheaper products [4], [5], [6]. Either way the die ships as a lower-core-count part. We call this *silicon harvesting*, or *die harvesting*, to distinguish it from the now more common architectural sense of "core harvesting" as lending idle cores *temporarily* to co-located work [7]; the two are unrelated. The practice is openly acknowledged, but the resulting *per-die* maps of which units were disabled are not published: a buyer is told the part has 16 cores, not *which* 16, and the fleet-wide readouts we know of ran code on each node [5], [8]. Those maps carry information about a vendor's defect and binning process and about a fleet's provenance that vendors do not publish.

Independently, HPC operators have begun publishing holistic out-of-band monitoring at fleet scale [9]. Such telemetry carries an unintended side channel: a baseboard management controller exposes one thermal sensor per core position on the die and does not renumber the surviving cores compactly, so the sensors that never report mark exactly the disabled positions, the die's harvest map. The owner of one POWER9 machine can read this from its sensors [10]; published fleet telemetry discloses it for every die at once.

We exploit this to recover harvest maps for 1,962 of the 1,968 processor sockets of CINECA's Marconi100, a Top-10 system, from a public dataset, and to track them across 858 days. Our contributions are:

- A recovery method (Sec. III) with an explicit validity argument, needing only per-core sensor telemetry that operators already collect, **validated against the operating system's and the scheduler's own core counts** (Sec. III-B).
- To our knowledge the first characterisation of harvest maps recovered from *published* out-of-band telemetry, without access to the machines: fleet-scale, and tracked daily over 31 months. On these 16-core parts every disabled core lies in a fully disabled POWER9 slice, maps are per-die rather than per-SKU, and harvested slices cluster, above all within the quads IBM documents.
- Evidence of rack-aligned *lot structure* from telemetry alone: a sharp changepoint, predictable for held-out racks, separating two populations of parts with markedly different binning statistics, most plausibly two procurement lots.
- A longitudinal result over 1.67 M socket-days: maps are near-static, configuration changes coincide with reboots **measured against an independent boot signal** rather than inferred from reporting gaps (Sec. IV-H), and field core-guard episodes are isolated, the nine in steady state each matched by the OS's own CPU count.
- A taxonomy of map changes by the event each is consistent with (Sec. IV-I), which also traces processors moved between nodes and so separates a 2.1 % annual

die-change rate from an inferred 1.0–1.2 % rate of new processors, with no asset database.

- Two bounded nulls that matter operationally: an equivalence analysis bounding the map's association with socket power to a few watts, below the spread between sockets (Sec. IV-L), and a survival model finding no useful prediction of field replacement (Sec. IV-M).
- A quantification of the disclosure itself (Sec. IV-N): a single 20 s sampling interval recovers the exact map for 99.95 % of socket-intervals, and neither reducing resolution nor random deletion short of destroying the data mitigates it.

II. Background

A. The machine, from room to core

Marconi100's 980 compute nodes occupy 49 rack cabinets of 20 nodes each, arranged in three rows of the machine room, with $\mathrm{id} = 20\,\mathrm{rack} + \mathrm{slot}$ and every rack's (X, Y) floor position documented [11]. Each node is an AC922 8335-GTG chassis carrying *two* CPU sockets and four V100 GPUs. Each socket holds one POWER9 die; each die carries 24 SMT4 core positions organised as 12 slices; each slice is two cores plus their shared L2 and L3. Fig. 1 shows the chain. Two consequences matter throughout: the unit we recover a map for is a *socket*, two per node rather than one, and the unit disabled on these parts is a *slice*, not a core. Sec. IV-A fixes the exact socket populations.

B. POWER9 slice organisation

The POWER9 die used in the IBM Power System AC922 contains 24 SMT4 cores (equivalently 12 SMT8 cores) on a $695\,\mathrm{mm}^2$ die in 14 nm FinFET with 17 metal layers [12]. Critically, *IBM groups each pair of SMT4 cores with a 512 kB L2 and a 10 MB L3 into a processor-pair cache slice* [6, pp. 30, 155], an EX in its firmware [13]. We call this unit a *slice*; IBM also uses the word for the execution slices inside a core [14, slide 7], which we never mean. IBM's Hot Chips 28 presentation shows the 24 cores above 12×10 MB L3 blocks, attached to a 7 TB/s on-chip fabric at $256\,\mathrm{GB/s} \times 12$ [14, slides 4, 11]. Two slices form a *quad*. A bad L2 disables both cores of its slice [15, pp. 263–264], but a single core can also be deconfigured [6, p. 311], and an 8-core POWER9 part runs one core in each of eight slices [10]; whether these parts disable whole slices is therefore an empirical question (Sec. IV-A).

Marconi100 nodes are AC922 model 8335-GTG with $2\times$ 16-core POWER9 at 2.6/3.1 GHz and $4\times$ NVIDIA V100 [16], [17], [18]. This model uses the 24-core scale-out POWER9 die in 16- and 20-core versions [16], so **a 16-core part is a 24-core die with eight cores disabled**; which eight is the object of our study. IBM documents two routes to a disabled core [6, p. 311]: at each boot the firmware deconfigures cores with manufacturing defects, read from partial-good data in the module's VPD [13], and it can deconfigure working cores not licensed for use, e.g. through a factory core-count override [19]. Telemetry shows *which* positions are disabled, not which route disabled them. Fig. 2 shows the organisation annotated with the harvest rates we measure.

C. The M100 ExaData dataset

M100 ExaData [9] publishes 49.9 TB of telemetry from Marconi100 covering 2020-03 to 2022-09 across 980+ nodes; we use all twelve monthly-archive releases [20]. The IPMI plugin collects BMC sensor data at 20 s cadence. The documentation describes the core-temperature family as `pX_coreY_temp`, "Temperature of core n. Y in the CPU socket n. X. X=0..1, Y=0..23" [11]. *This specifies the BMC sensor-ID namespace, not the active core count*. That distinction is the entry point for this study.

III. Method

A. Recovering the disable map

For each month we enumerate the `metric=pX_coreY_temp` partitions and count samples per (node, socket, core). A core index is *present* for a socket if it emits any sample in the window. The disable map of socket s is $M_s \in \{0,1\}^{12}$ over slices $k = \lfloor \mathrm{core}/2 \rfloor$, with $M_{s,k} = 1$ when neither core of slice k reports (the slice is disabled); every socket analysed has $\sum_k M_{s,k} = 4$. We call M_s the socket's *harvest map*; it records which positions are disabled, not why.

The key validity argument is that **the BMC does not renumber active cores compactly**: were that so, every socket would report indices 0–15. Instead we observe arbitrary 16-element subsets of $\{0, \ldots, 23\}$, so the index is fixed to the core rather than to its rank among the survivors; IBM's numbering ties it to a position on the die (Sec. IV-F).

B. External validation of the premise

Everything in this paper rests on one premise: that the per-core sensors that never report are exactly the die's disabled positions. The argument of Sec. III is internal: a renumbering BMC would yield one pattern, and we see 443. We can do better, because M100 ExaData carries two collectors that never touch the IPMI sensor family and that both know how many cores exist.

The operating system. The Ganglia plugin reports `cpu_num`, the node's own logical-CPU count. POWER9 runs four SMT threads per core, so two sockets of 16 cores should present $2{\times}16{\times}4 = 128$ logical CPUs. Across the whole record (**847,297,478 samples, 990 nodes, the 29 months in which Ganglia collected this metric**), the OS reports exactly 128 on **99.9853 %** of samples. Restricted to the single month 2022-08 used for the thermal work (40.9 M samples, 988 nodes) the figure is **99.997 %**. We give both because both appear below, and they are different windows rather than different measurements of the same one.

The scheduler. SLURM's configured-CPU totals across the four partitions are $110{,}208 + 12{,}672 + 2{,}560 + 256 = 125{,}696$, which is 982×128. The scheduler agrees, from a third independent path.

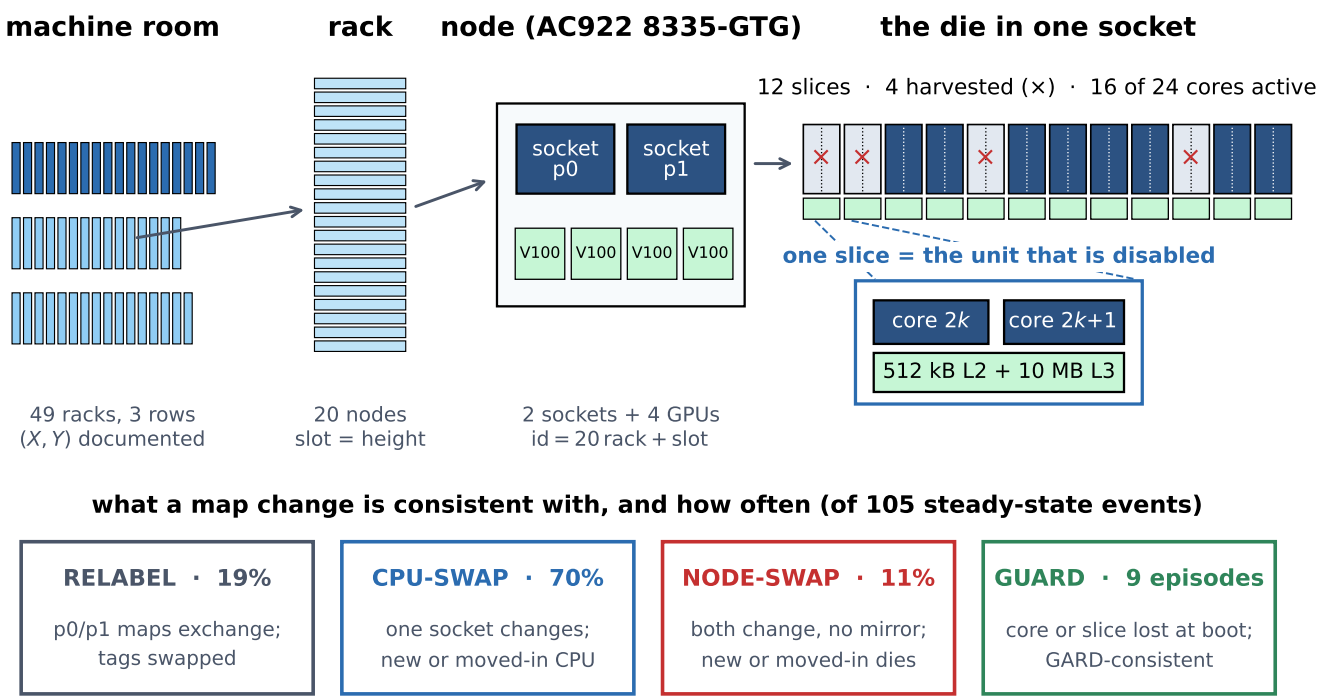


Fig. 1. The physical hierarchy this paper reasons over, and what a change in a recovered map means at each level. A socket's harvest map is carried by the *die currently seated in that position*, though it is not a serial number, so a change in the map is most plausibly a change of die, of position labelling, or of configuration. The bottom row names the event each class of change is consistent with (Sec. IV-I); GUARD counts the nine deconfigurations the OS confirms in steady state (Sec. IV-H).

The one exception is itself a confirmation. Exactly one node deviates *in the month 2022-08*: node 145 reports 124 logical CPUs, one core short, across 2022-08-11–12; over the same two days the BMC loses exactly one core sensor on socket p1 from a reboot, and the socket then returns with a different map (the slice at cores $\{22, 23\}$ replaced by $\{18, 19\}$) while socket p0 is unchanged all month. Two collectors with nothing in common flag the same node, and the same number of cores, in the same 48 hours. This ties the sensor-presence signal to real hardware configuration rather than to an artefact of the monitoring stack.

The premise is therefore externally confirmed at fleet scale for the *count*, by the OS and the scheduler independently; *which* positions are disabled rests on the non-compact indexing above and on Secs. IV-A and IV-F.

C. Pipeline, filters, and what each filter protects against

The recovery above is one line of logic; making it reliable at 1.67 M socket-days is mostly a question of filters, and we state them here rather than introducing them where they are first used.

Stage 1: extraction. Each monthly archive is scanned once and only the 48 `p{0,1}_core{0..23}_temp` partitions are extracted. A **completeness gate** refuses to emit a month unless all 48 are present. This is not defensive boilerplate: a partial extraction is indistinguishable, downstream, from a socket with disabled cores, and an early version of this pipeline produced exactly that artefact for three months (Sec. III-D).

Stage 2: daily aggregation. Samples are reduced to a count per (node, socket, core, day), giving a 15 MB table from a 400 GB scan. Every structural and longitudinal result in this paper is computed from that table; only the thermal analysis returns to the raw series.

Stage 3: the clean-day filter. A socket-day is *clean* when exactly 16 cores report *and* each reports more than 90 % of the 4,320 samples expected at 20 s cadence. This is the load-bearing choice in the paper, so its effect is stated plainly: it retains 1,517,473 of 1,671,297 socket-days (90.8 %), and Sec. IV-G shows the lot result is unchanged when the threshold is tightened to sockets with $\geq$90 % clean-day coverage. The filter exists because a partially-collected day mimics a configuration change. Because it conditions on a count of 16, it cannot reveal a configuration of another size; the census of every other socket-day in Sec. IV-H finds only short deconfiguration episodes and partially sampled days.

Stage 4: per-socket map. A socket is summarised by the modal harvest map over its clean days (median 775 days supporting each map). Sockets whose modal map does not have exactly four harvested slices are excluded; this removes 6 of 1,968. For the 206 sockets whose map changes, the modal map is that of one die, so time-dependent analyses use the map in force at the time.

Cost. One full pass is a 400 GB sequential scan, roughly two hours on a single node; re-running every result from the published derived table takes minutes.

D. Failure modes of the method

Four things could make this recovery wrong, and each has a detector.

Partial ingestion. If archive extraction silently truncates, absent metrics look like harvested cores. *Detector:* the 48-metric completeness gate. We found this the hard way: three months were once written from incomplete extractions, one of them missing all 24 sensors of socket p1, which would have read as a socket with no active cores at all.

Compact renumbering. If a BMC renumbered surviving cores $0..15$, the method would recover nothing but a constant. *Detector:* we observe arbitrary 16-element subsets of $\{0,\ldots,23\}$ and 443 distinct patterns; a renumbering BMC would produce exactly one.

Per-metric collector faults. A collector that drops individual metric streams mimics deconfiguration. *Detectors:* odd active-core counts, which no baseline map produces (39 socket-days; a single-core deconfiguration can, Sec. IV-H); reduced counts that are strict subsets of the socket's modal map; and, for the guard episodes specifically, the control that every other sensor on the same socket kept full cadence (Sec. IV-H).

Socket-tag swaps. If the collector exchanges the p0/p1 tags, a longitudinal study silently mixes two dies. *Detector:* the mirror test of Sec. IV-I, which identifies 19 % of steady-state map changes as relabelling, which moves no die.

E. Null model for clustering

Every socket has exactly four harvested slices, so the constraint induces negative dependence and a naive independence null is invalid. We use the curveball algorithm [21], which preserves *both* row sums (4 per socket) and column sums (per-slice marginals) while randomising association structure, and assert both on every draw. The $B = 100{,}000$ draws are exchangeable with the data by Besag and Clifford's parallel construction [22] (40 global curveball trades [23] to a hidden matrix, then 40 more for each draw), so the Monte-Carlo $p = (b{+}1)/(B{+}1)$, where b draws are at least as extreme, is valid however slowly the chain mixes. A z against a simulated null is a Monte-Carlo standardised distance, $(T - \bar{T}_{\mathrm{null}})/\mathrm{SD}_{\mathrm{null}}$; significance is read from p.

F. Thermal probing of die topology

Cores on a socket share workload and inlet air, producing a dominant common mode. We remove it by subtracting the per-timestamp cross-core mean, double-centre, and compute residual correlations at the sensors' native 20 s cadence. No resampling is applied, and Sec. IV-E shows why that matters. Cores that are neighbours on the die, or share a voltage and frequency domain, should retain positive residual coupling.

Caveat: Common-mode removal forces each row of the residual matrix to sum to approximately zero, so negative values at large index gaps are partly mechanical. We therefore rely on contrasts *at identical index distance* and on ordering tests, which the constraint cannot bias: it treats every core alike, so it can rescale an ordering but not create one. Every thermal interval resamples the 49 racks.

IV. Results

A. Every socket is 16 cores; granularity is the slice

Populations: Two socket populations recur and are kept distinct throughout: $\mathcal{S}_{\mathrm{all}}$, the 1,968 sockets appearing anywhere in the record (984 nodes), and $\mathcal{S}_{\mathrm{map}}$, the 1,962 with at least one clean day and hence a well-defined harvest map. All structural results below are over $\mathcal{S}_{\mathrm{map}}$ and the full 31-month record. One further distinction matters only for the rack-level analyses: node identities run to 983, four beyond the 980 that fill the 49 documented cabinets, and those four have no documented (X, Y) floor position. The 2 sockets of theirs that carry a map are excluded wherever a result is indexed by rack, leaving 1,960; being a four-node group at the end of the id range, they would otherwise form a degenerate segment that captures the argmax of any unguarded changepoint scan (Sec. IV-G).

TABLE I
DISABLE-MAP STRUCTURE OVER THE FULL 31-MONTH RECORD.

Sockets with a well-defined harvest map	1,962
Median clean days supporting each map	775
Socket-days reporting exactly 16 cores	**99.9847%**
Fully-sampled socket-days with >16 cores	0
Disabled cores form complete $(2k, 2k{+}1)$ pairs	**1,962 (100%)**

The perfect pair structure (Table I) is the central structural finding: it matches the POWER9 slice exactly, and no socket's map disables one core of a slice and keeps the other, although the hardware can deconfigure single cores [6, p. 311] and an 8-core POWER9 part runs one core in each of eight slices [10]. Slice granularity is therefore a property of these 16-core parts, not a constraint of the die; the 4,200 Xeon Phi 7250s of another centre likewise lose whole two-core tiles [5].

B. The map is per-die, not per-SKU

443 distinct patterns appear among the $\binom{12}{4} = 495$ possible (Fig. 8). Per-die means that the map varies from processor to processor rather than being fixed by the SKU; POWER9 firmware reads each module's partial-good record from that module's own vital product data at boot [13]. The most common map (slices 0–3) covers only 5.5% of sockets. Within a node the two sockets match in just 1.2 % of cases (12 of 981 nodes), with mean overlap 1.641 slices against a shuffled null of 1.470 ± 0.026 ($z = +6.6$), so they are nearly independent, with a small excess consistent with same-lot pairing.

C. Marginals are strongly non-uniform

Slice 0 is harvested on 61.2% of sockets against a uniform expectation of 33.3% ($\chi^2 = 807.2$, $\mathrm{df} = 11$, $p \ll 0.001$,

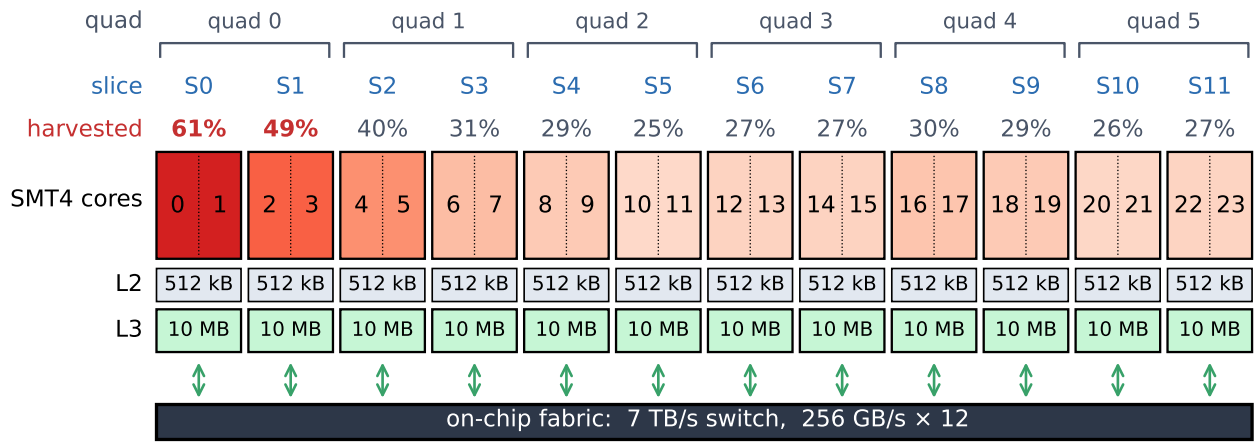


Fig. 2. POWER9 slice organisation (after [14, slides 4, 11]) annotated with the measured per-slice harvest rate across the 1,962 Marconi100 sockets with a well-defined map. Each slice holds two SMT4 cores sharing 512 kB L2 and 10 MB L3 on a common fabric port. In every recovered map, disabled cores come in whole slices. The horizontal ordering is the IPMI sensor index, drawn as one row for clarity; brackets mark the six quads, which on the die sit in two bands of three (Sec. IV-F).

conservative since each socket contributes four distinct slices; a Monte-Carlo test with that constraint gives $p = 1/(B+1) = 5 \times 10^{-5}$ and a rack-clustered Wald test $p = 3 \times 10^{-6}$); Fig. 3 splits this by lot. These marginals are stable in time: across the 30 months with clean data the largest standard deviation of any slice's monthly rate is **0.25 percentage points**, and slice 0 ranges only 60.9–61.8% (Fig. 12). Changing composition does not produce this: a fixed cohort of the 1,900 sockets with a clean day in every post-commissioning month has a largest standard deviation of 0.22 points, and only 0.21 % of its socket-months change map from the month before.

D. Harvested slices cluster, above all within quads

Under the curveball null the mean index gap between harvested slices is 4.242 versus 4.582 ± 0.016 ($z = -20.7$; none of the 100,000 null draws is as small, $p = 1/(B+1) \approx 10^{-5}$). Of the $\binom{12}{2} = 66$ pair counts, 20 are significant at family-wise error 0.05, identically under a Monte-Carlo Bonferroni test ($p \leq 0.05/66$) and a max-$|z|$ test (Fig. 4).

Nine are positive excesses and eleven deficits. *Exactly eight* positive excesses have $|\Delta k| \leq 2$: the six pairs $(2m, 2m+1)$, which share a *quad*, POWER9's group of four cores whose voltage and frequency change together [6], plus 1&2 and 0&2. The most significant is 0&1 ($z = +16.1$); the ninth excess, 7&10, is at $|\Delta k| = 3$. Of the five index neighbours that straddle two quads only 1&2 shows an excess; 3&4, 5&6, 7&8 and 9&10 sit at 0.89–$1.09\times$ the null. Nine of the eleven deficits are at $|\Delta k| \geq 4$, the exceptions being 2&4 and 1&4. A 10,000-draw run of the Besag–Clifford sequential null (Sec. III-E) reaches the same 66 decisions. Part of the pattern is lot mixture: with trades restricted to within each lot, so each keeps its own marginals, the mean-gap z is -15.4 and 11 pairs remain, seven of the eight near excesses (0&2 falls to $z = +3.1$) and four deficits, all at $|\Delta k| \geq 6$.

The six same-quad pairs are also neighbours on the die. IBM's chip diagram places the two slices of each quad next to each other and consecutive quads in opposite bands [15, Fig. 2] (Sec. IV-F), so this part of the clustering is spatial, at the scale of a quad; the other two near excesses, 1&2 and 0&2, join slices in opposite bands of one die column. A pairwise model with separate adjacency terms inside and across quads puts most of the adjacency inside the quad ($\gamma = +0.92$ against $+0.28$, Sec. IV-J). Spatially clustered defects [1], [24] and disabling at quad granularity would both produce this, and the telemetry cannot tell them apart (OI-3).

How big, not just how significant: A z is a signal-to-noise ratio and grows with $\sqrt{n}$, so the values above report sample size as much as effect. Fig. 4 therefore plots the effect itself, $\log_2$ of observed over expected co-harvest, against slice distance. The excess is real but moderate: averaged over pairs at $|\Delta k| \leq 2$ it is $+0.165$ ($1.12\times$ the null), against -0.089 ($0.94\times$) for more distant pairs. The six largest excesses, 1.32–$1.73\times$ (strongest 6&7), are exactly the six same-quad pairs. Clustering of this size is easy to detect across 1,962 sockets and easy to overstate from a z alone, so both are given.

E. Thermal coupling follows the quad, not the slice

We probe **every** socket with both halves fully configured, **1,948 sockets across 974 nodes**, over the full 31-day month 2022-08 (Fig. 5). At *identical* index distance of 1, within-slice siblings couple more strongly than cross-slice neighbours on average, $r = +0.372$ against $+0.312$ (15,584 and 10,122 pairs), a contrast of $+0.061$ (95 % CI $[0.047, 0.076]$, resampling the 49 racks). That average mixes two kinds of neighbour. IBM's chip diagram stacks the four cores of each quad in one column and alternates consecutive quads between the die's two bands [15], so index neighbours in different quads are not neighbours on the die. Cross-slice neighbours within a quad

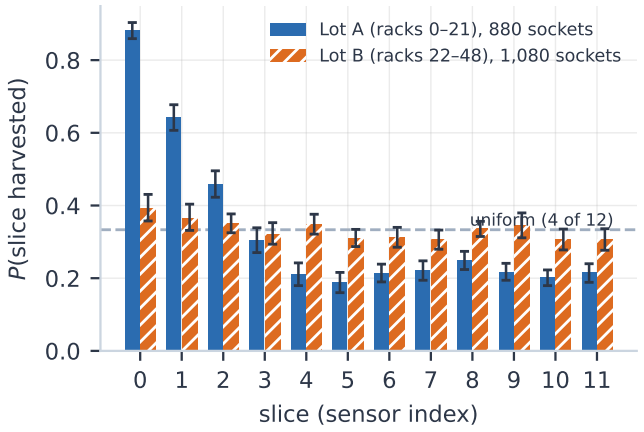


Fig. 3. Per-slice harvest probability by lot (880 and 1,080 sockets in racks 0–21 and 22–48), with rack-clustered bootstrap 95 % intervals (2,000 resamples of the racks within each lot). Lot A falls steeply with slice index; Lot B is nearly flat.

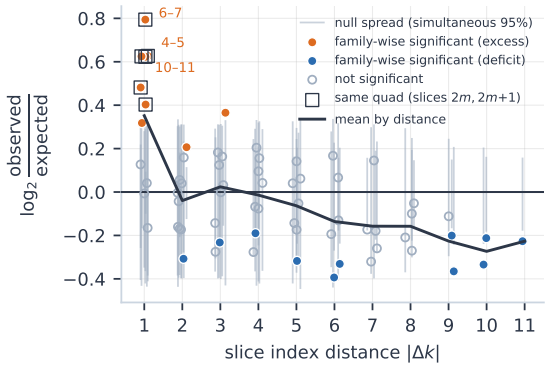


Fig. 4. Co-harvest effect size against the curveball null, $\log_2$ of observed over expected, for all 66 slice pairs over 1,962 sockets, against their sensor-index distance. The excess is confined to near-adjacent pairs, largest for the six same-quad pairs (squares), and reverses at distance. Grey bars are the simultaneous 95 % band of the null (its spread, not a confidence interval); filled points are family-wise significant against 100,000 null draws. Plotting the ratio rather than a z keeps sample size out of the vertical axis.

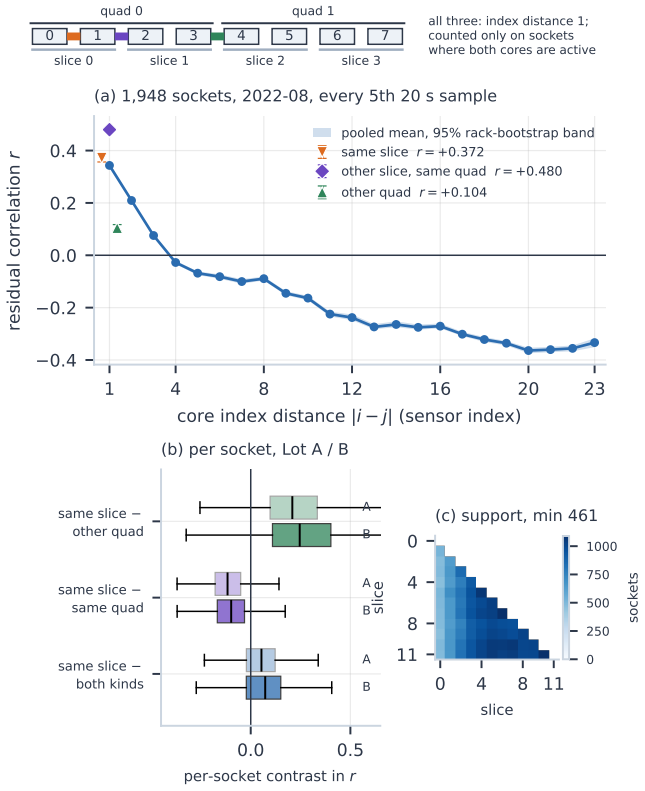


Fig. 5. Residual thermal coupling over the 1,948-socket census. (a) Correlation against sensor-index distance, band a 95 % rack-bootstrap interval; at distance 1 the three kinds of neighbour are shown apart. (b) Per-socket, position-matched contrasts by lot. (c) Sockets supporting each slice-pair cell. Top: the pair-inclusion rule.

(55 % of them) couple at $r = +0.480$, those across a quad boundary at $+0.104$. Matched within each socket, siblings couple *less* than the adjacent core of the other slice in their quad (-0.102, CI $[-0.114, -0.089]$) and far more than cores across a quad boundary ($+0.246$, $[0.227, 0.267]$); both keep their sign in each lot, socket position and month tested, but their mixture does not: it is -0.004 ($[-0.019, 0.013]$) in 2022-02 and $+0.104$ in 2021-05. The thermal unit is the quad; the slice is the unit of disabling.

A census, not a sample: Subsampling in *time* rather than in nodes lets the whole fleet be processed in one pass: we keep every fifth 20 s sample, which lowers the sample count without smoothing and so, unlike a coarser alignment grid, does not inflate correlations. The layout score of Sec. IV-F moves from -0.787 on the 180-socket subsample of Table II to -0.791 fleet-wide. Every thermal number reported here is a fleet-wide measurement, and every interval resamples racks: treating the pair correlations as independent, as an earlier version did, understates the variance of the contrast about five-fold.

The effect size depends on temporal resolution; the ordering does not: Cores are sampled every 20 s and their timestamps coincide across a socket, so no resampling is required: an inner join at native resolution retains 133,907 of 133,920 possible rows (99.99 %). Earlier versions of this analysis nonetheless aligned onto a one-minute grid, a step introduced to work around an unrelated bug and never revisited. It is not neutral (Table II): coarser averaging suppresses high-frequency noise and inflates every correlation, within-slice more than cross-slice, so within this subsample the contrast grows from $+0.063$ at 20 s to $+0.140$ at 15 min. **We therefore quote the native-resolution figure, which is the most conservative.** What does *not* move is the ordering (within-slice exceeds cross-slice at every grid) and the layout score of Sec. IV-F, which sits at $-0.787, -0.773, -0.766, -0.767$ across the four grids.

What couples the cores?: Inside a quad, physical adjacency and a shared operating point coincide: its four cores are neighbours on the die [15] and change voltage and frequency together [6], so their power, and hence their temperature, can

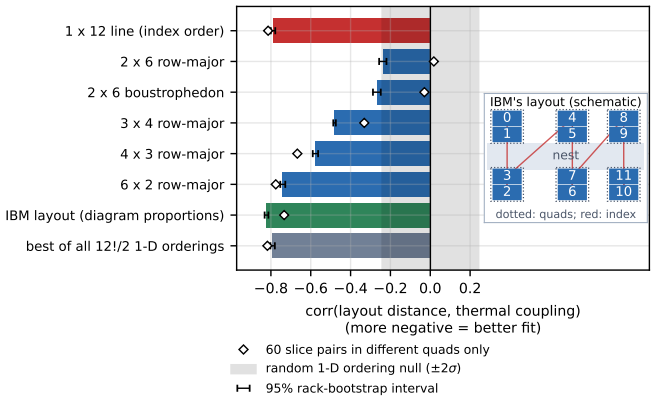


Fig. 6. Candidate slice layouts scored against thermal coupling (Pearson correlation over the 66 slice pairs; more negative = better), with 95 % rack-bootstrap intervals; diamonds: the same score over the 60 pairs in different quads; band: random one-dimensional ordering null $\pm 2\sigma$. Green: IBM's documented layout at the diagram's proportions (inset: dotted boxes are quads, red is the index order), ahead only through its same-quad pairs. The index order (red bar) is second of all $12!/2$ orderings. The layout itself is documented, not inferred: the scores are descriptive.

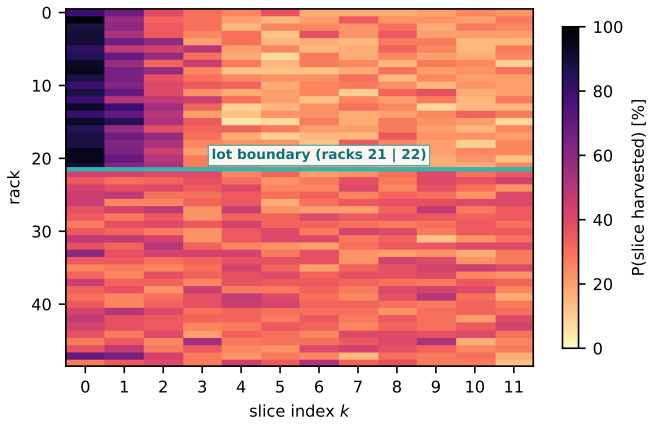


Fig. 7. Harvest probability by rack (rows: the 49 documented cabinets, 20 nodes each, in node-id order) and slice (columns: slice index). One rack separates two regimes: Lot A concentrates harvesting in slices 0–2 (slice 0: 88 % against 39 %) while Lot B spreads it, taking slices 4–11 some 9–14 points more often; only slice 3 is unchanged. Racks 21 and 22 are adjacent cabinets in the same row, and Sec. IV-G tests the room directly.

move together without any heat exchange. Lagged correlations cannot separate the two. On-die hot spots evolve over milliseconds [25], far faster than the 20 s sample, while the heat sink responds over minutes; at true 20 s lags every kind of neighbour loses roughly a quarter to a half of its lag-0 coupling within one sample (24 % within a slice, 52 % across quads) and then changes by less than 0.02 out to 200 s.

F. Sensor index and position on the die

Is the sensor index merely a label, or does it track physical position? IBM documents the answer. Its chip diagram labels the core chiplets EC00–EC23 [15, Fig. 2, p. 33], the core numbers that the per-core sensors carry [10], and places the six quads in two bands of three, each quad's cores stacked

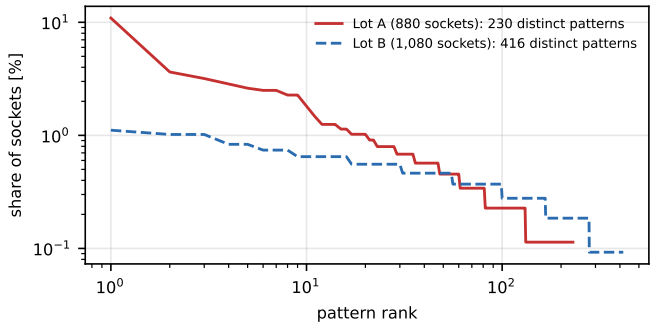


Fig. 8. Rank–frequency of disable patterns, as a share of each lot's sockets (880 and 1,080 in racks 0–21 and 22–48). Lot B is markedly more diverse: 458 ± 20 effective maps against 138 ± 9 (2^H, Chao–Shen entropy, rack-jackknife SE; whole-rack permutation $p = 5 \times 10^{-4}$). Chao–Shen corrects for patterns not yet seen, so the effective count can exceed the 416 and 230 observed.

TABLE II
EFFECT OF THE ALIGNMENT GRID ON THE SLICE-LEVEL CONTRAST.

Grid	rows/socket	within	cross	1×12 fit
20 s (native)	**133,907**	**+0.377**	**+0.314**	−0.787
1 min	44,640	+0.411	+0.329	−0.773
5 min	8,928	+0.466	+0.351	−0.766
15 min	2,976	+0.508	+0.368	−0.767

This table is a 180-socket subsample, held fixed across the four rows so that only the grid varies. Its 20 s row therefore differs slightly from the 1,948-socket census reported in the text (+0.372/+0.312, fit −0.791). The point of the table is the trend down the rows, not the absolute values. The slice-level contrast is a mixture of same-quad and cross-quad neighbours, as this subsection shows; it is tabulated here only because it is the quantity earlier drafts reported, so that the effect of the grid on it is visible.

in one column (Fig. 6, inset), so index neighbours are die neighbours only within a quad. What can the thermal data add? A candidate layout's score is the Pearson correlation, over the 66 slice pairs with equal weight, between its slice distance and the slices' thermal similarity (the mean of their four cross-slice core-pair correlations); coupling falls with distance, so a better layout scores more negative.

Over all 1,948 sockets the index order scores −0.791 (95 % CI $[-0.800, -0.778]$, resampling racks; random-ordering null -0.003 ± 0.123, $z = -6.41$, rack-bootstrap $[-6.55, -6.24]$) and beats every grid of unit spacing (Fig. 6). An exact search over all $12!/2$ one-dimensional orderings (dynamic programming over the 2^{12} prefix sets) finds $[0, \ldots, 9, 11, 10]$ at −0.793; the index order ranks second of 239,500,800 in each of three months. IBM's layout, at the diagram's proportions, scores −0.824 ($[-0.832, -0.813]$), though that score turns on the diagram's assumed proportions rather than on sampling: over 90 column, gap and pitch scalings it ranges −0.846 to −0.311 and beats the best one-dimensional ordering in 34 of them, and in 2022-02 the two tie exactly. And it wins only through the six same-quad pairs: over the 60 pairs in different quads the index order fits better (−0.814 against −0.734).

What the coupling follows: The score uses cross-slice pairs only, so shared L2/L3 does not enter. Within a quad

the coupling matches IBM's layout: at index distance 1, neighbours in one quad couple at $+0.37$ to $+0.48$, neighbours in different quads at $+0.10$. Beyond the quad it follows the order in which the surviving cores are numbered, not the diagram. Within a die column, slice $4c{+}1$ couples on average more with its index neighbour $4c{+}2$ at the far die edge than with the nearer slice $4c{+}3$ across the nest (by 0.055, CI $[0.046, 0.066]$), where heat-source models on IBM's layout predict the opposite sign; and for the same two slices, coupling falls by 0.080 ($[0.073, 0.086]$) per active slice between them, against at most 0.032 in those models. All three months agree. Only a small same-band excess ($+0.044$, $[0.040, 0.049]$) points to the diagram's two bands. **The layout therefore comes from IBM's documentation, not from telemetry: thermal coupling confirms the quads and, beyond them, follows the core numbering,** plausibly through work placed on consecutive logical CPUs. The spatial reading of Sec. IV-D rests on the documented quads, not on a thermally recovered ordering.

G. A lot boundary at rack 22

What a rack is here: Nodes fill 49 cabinets of 20 with $\mathrm{id} = 20\,\mathrm{rack}{+}\mathrm{slot}$, and every cabinet's (X, Y) position on the machine-room floor is documented [11], so the rack index is a physical coordinate, not a label. That is why it is informative, and why it must be handled carefully: it conflates the *hardware population* (parts installed together would fill neighbouring racks) with *room position* (cooling, airflow, height in cabinet). We separate the two below. We call the two rack-aligned populations *lots*; procurement batches are their most plausible reading, but no delivery record was available to confirm it.

A changepoint scan maximises Welch t at rack 22, $t = 26.6$ (Fig. 7). At node granularity the boundary is sharper still: $t = 26.8$ between nodes 438 and 439, with the 40 sockets immediately before at 95.0 % slice-0 harvested and the 40 immediately after at 42.5 %. Node 439 sits in the topmost slot of rack 21 yet behaves like the later population (0 of its 2 sockets have slice 0 harvested, against 2 of 2 for each of nodes 430–438), so the boundary is rack-aligned to within a single node (Table III).

TABLE III
TWO POPULATIONS EITHER SIDE OF THE LOT BOUNDARY, AT THE TWO GRANULARITIES THE BOUNDARY CAN BE DRAWN AT.

Cut	Population	Sockets	P(slice 0)	Patterns
rack	Lot A (racks 0–21)	880	88.2%	230
	Lot B (racks 22–48)	1,080	39.2%	416
node	Lot A (nodes 0–438)	878	88.4%	229
	Lot B (nodes 439–979)	1,082	39.1%	416

The two cuts differ by exactly one node: the rack cut places node 439, the topmost slot of rack 21, in Lot A, whereas the node-level changepoint assigns it to Lot B, where its harvest map says it belongs. Nothing downstream turns on the choice; we use the *rack* cut throughout, because the lot-level model of Sec. IV-J resamples racks. Node 980's two sockets have no documented cabinet (Sec. IV-A) and are excluded from both cuts, as in every rack-indexed result; including them gives 1,082 and 1,084 and moves no rate by more than 0.1 points.

Lot A marginals fall steeply with slice index; Lot B is nearly flat. Beyond the boundary the rack carries little: same-rack sockets share more harvested slices than a fleet-wide shuffle predicts (1.565 vs 1.453 ± 0.008, $z = +13.3$), but shuffling within each lot removes the excess (1.559 ± 0.007, $z = +0.9$), and between-rack variance within lots is at most marginal (smallest $p = 0.017$, slice 0, of 12 slices).

The boundary, tested rather than maximised: A t of 26.6 obtained by scanning every split is a *maximum* over dependent tests, and cannot be read as a significance level. We therefore calibrate it. Permuting whole racks (which preserves rack sizes and any within-rack dependence), the null distribution of $\max|t|$ has mean 7.08 ± 2.41 and never exceeds 17.95 in 4,000 draws, against the observed 26.6 (Fig. 9): calibrated $p = 2.5 \times 10^{-4}$, the floor of a 4,000-draw test. Resampling sockets or whole nodes within each rack, with the racks held in place, puts the breakpoint at rack 22 in 100 % and 99.6 % of 2,000 resamples: a stability check, not a confidence interval. Under the fitted two-lot model, resampling whole racks within each side returns rack 22 in 99.7 % and a rack-level beta-binomial parametric bootstrap in 98.7 %, and the rack-level likelihood favours rack 22 over rack 23 by a factor of 5.6×10^4. Leaving out each rack in turn, **all 49** fits split the remaining racks where the full data do. The scan is restricted to boundaries leaving at least 100 sockets per side; without that guard the argmax is captured by a four-node partial rack at the end of the id range, which we exclude and report (Sec. IV-A).

Two populations, chosen out of sample: Rather than a stopping rule on $|t|$, we segment the rack-ordered fleet into k contiguous populations by dynamic programming and choose k by *rack-held-out* likelihood, since in-sample fit always rewards more segments. Under a twelve-slice Bernoulli score, held-out log-likelihood per socket is -7.360 at $k = 1$, -7.133 **at** $k = 2$, then -7.139, -7.157 and -7.149 at $k = 3, 4, 5$. Because that score ignores the four-slice constraint, we repeat the selection with three models on the 495 valid maps (conditional Bernoulli, the pairwise model of Sec. IV-J, a Dirichlet-smoothed multinomial): each selects $k = 2$ with the cut at rack 22, except the multinomial at its lightest smoothing, the weakest predictor of all, which prefers $k = 1$. For slice 0, a step also predicts held-out racks better than a logistic trend in rack index. Among one to five rack-ordered populations, two is the out-of-sample optimum, and the change is a step rather than a gradient.

The lot is predictable, not merely detectable: The operationally meaningful question is whether a harvest map places a rack the model has never seen in the right lot. Holding out each rack in turn, we re-locate the boundary and fit the two lot models on the other 48 racks only, then classify the held-out rack's sockets by likelihood ratio: **72.9 % of individual sockets** are assigned to the lot the full data give them, against a 55.1 % majority-class baseline, and **47 of 49 held-out racks (95.9 %)** by majority vote. Lot membership is thus predictable out of sample, which a large in-sample t alone would not establish; the lots themselves remain inferred, not documented deliveries.

The hardware, not the room: Because rack index conflates hardware population with geometry, we test the geometric alternative directly. Raw correlations with slice-0 status are substantial for the machine-room row ($r = -0.414$ for Y) and visible for aisle position ($r = +0.124$), exactly as expected when both track rack index. **Once the lot is accounted for, they vanish**: partial correlations are $+0.017$ for Y ($p = 0.44$) and $+0.014$ for X ($p = 0.53$). Only the slot, meaning height within the cabinet, retains a residual -0.054 ($p = 0.017$, 0.3 % of variance), which is marginal under a three-test correction and would in any case be consistent with racks being filled in installation order. The boundary also falls inside a row: in the middle row, Lot A racks 18–21 have slice 0 harvested on 91.9 % of sockets and Lot B racks 22–32 on 39.1 % (rack rates 87.5–95.0 % against 27.5–47.5 %), and adjacent cabinets 21 and 22 differ 90.0 % versus 42.5 %. Nor does the signature stay with the rack: dies arriving after commissioning in replacement-class map changes (Sec. IV-I) have slice 0 harvested on 33 of 54 in Lot A racks (61 %, node-bootstrap 95 % CI $[48, 73]$, against Lot A's 88 %) and on 21 of 42 in Lot B racks, with no detectable difference between the lots ($p = 0.30$). The signature belongs to the parts installed at commissioning, or to how they were set up then, not to the room or to any setting that stays with the rack. Rack order can still stand for how the two populations differed when they were racked (delivery, vendor-side provisioning or installation-time setup), which telemetry alone cannot separate.

The boundary is not an artefact of the coverage filter: Because every structural result is computed on *clean* socket-days, a lot-correlated dropout could in principle manufacture this contrast. It does not. Clean-day rates are near-identical between lots (0.909 vs 0.902), coverage is uncorrelated with the harvest pattern (corr with slice-0 status $= +0.004$, $p = 0.88$), and restricting to the 1,847 sockets with $\geq$90 % clean-day coverage leaves the contrast unchanged at $+49.0$ percentage points versus $+48.9$ over all sockets.

H. Configuration lifetime

POWER9 firmware *could* also alter the active set after manufacture: a core with a detected hardware error can be recorded in a persistent GARD record that hostboot applies at every IPL and `opal-gard` can clear, and a core-count override can switch cores off for licensing [6], [13], [19], [26]. Whether harvest maps are static in practice is therefore an empirical question, which we answer by aggregating every core sample to (node, socket, day) counts across the dataset. All numbers in this section are over the complete 31-month record (2020-03 to 2022-09).

The count is very nearly invariant: Across all 31 months (1,671,297 socket-days over 858 reporting days and 984 nodes), **99.9847% report exactly 16 active cores**. Of the 256 exceptions, 39 have an odd count, which no baseline map produces: 11 fall on days the OS confirms a single-core deconfiguration (below), 4 on one socket before OS collection began, 21 are one sensor lost while the OS counted all 128 CPUs, so the collector can also drop a core, and the remaining three are two unions spanning a reconfiguration and one socket with no earlier clean map. The 209 counts above 16 are all partially sampled days spanning a reconfiguration.[1] Only **nine** socket-days are fully-sampled with an even count below 16, and these are the subject of the guard analysis below.

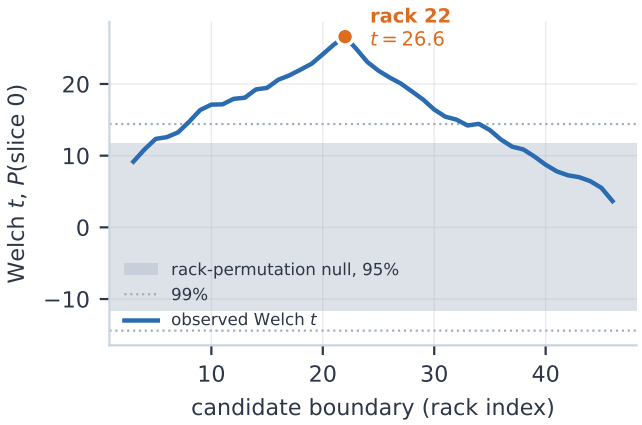


Fig. 9. The lot boundary against a simultaneous null. Welch t for P(slice 0) at every admissible boundary; shaded and dotted: the 95th and 99th percentiles of $\max|t|$ under 4,000 whole-rack permutations, a band for all boundaries at once. The observed peak at rack 22 ($t = 26.6$) lies far outside a null that never exceeds 17.95 (calibrated $p = 2.5 \times 10^{-4}$), so the boundary is not an artefact of scanning many splits.

Configurations are near-static: Restricting to *clean* socket-days (16 cores, each at >90% of the 4,320 samples expected at 20 s cadence) leaves 1,517,473 socket-days, a median of 775 observed days per socket. **89.5% of sockets never change configuration** (95 % CI $[88.1, 90.8]$) (1,756 of 1,962); 183 have two distinct sets, 19 have three, three have four and one has five.

Core-guard episodes: The nine anomalous socket-days resolve into **three sustained episodes on three distinct sockets**, the episodes that meet this strict criterion (Table IV). In every case the socket loses *exactly one slice*, a complete $(2k, 2k{+}1)$ pair, and runs on 14 cores for 2–4 consecutive fully-sampled days, a loss with the signature of a GARD record applied at IPL (the record itself is not in the telemetry). In the commissioning-era case the slice later returns; in both steady-state cases the socket instead returns with a different map (Table IV).

Rates need care. One of the three episodes (347 p1, 2020-03-13–16) falls inside the commissioning window identified and excluded later in this section, so the strict criterion gives **two steady-state episodes**. An episode counts only on fully-sampled days, so the denominator is *observed* exposure, 3,991 socket-years over the steady state: **5.0 strict episodes per 10,000 socket-years**, exact Poisson 95 % CI $[0.6, 18.1]$, a factor of thirty wide. Counting every deconfiguration the OS confirms, slice or single core (below), gives 26 $[12, 49]$ per

[1]The three groups quoted here are not a partition of the 256: two odd-count socket-days are also above 16 (17 and 23 active cores), and the tenth 14-core day is not fully sampled, so 39, 209 and nine cover 255 of them.

10,000 over 3,500 OS-observed socket-years. Both are order-of-magnitude statements, and durations are lower bounds.

Independently confirmed by the operating system. Of 990 nodes, exactly **three** ever report **120** logical CPUs, which is $128 - 8$, precisely one slice of two cores at four threads. Those three nodes are **347, 411 and 946**: the same three, and only those, on which the strict criterion finds a slice episode. Under a null in which the OS flagged three arbitrary nodes, the chance of matching our three is $1/\binom{990}{3} = 6.2 \times 10^{-9}$.

The dates agree too, and agree in the direction the method predicts. For node 411 the OS window is 2021-11-12–16 against a BMC episode of 11-13–15; for node 946, 2020-08-30–09-02 against 08-31–09-01. In both cases the OS window *brackets* the BMC one, exactly as it should, since a BMC episode is counted only on fully-sampled days and its duration is therefore a lower bound. Node 347's OS deviation is a recurrence: in 2020-10 the BMC loses the same slice again, on days too partially sampled for our strict criterion.

Three events recovered from thermal-sensor presence alone are confirmed, node-for-node, by the operating system's own CPU count, a quantity produced by a different collector, from a different layer of the stack, that never touches a BMC sensor.

Every OS departure from 128 is seen by the BMC. Six further nodes report **124** logical CPUs ($128 - 4$, one SMT4 *core*), and on each the BMC loses exactly one core sensor from the same boot (whole days on five, from mid-day on node 145), a single-core deconfiguration our slice-granular criterion excludes. All **nine** OS departures are thus matched node for node and day for day, and each is followed within six days by a new map on that socket that no other socket gave up that week, as a failing processor replaced would produce (Sec. IV-I).

Not an artefact of a tight filter. The criterion above is strict, so we re-ran the search against the *adjacent observed day* rather than the modal map, requiring only that surviving slices exceed 50 % sampling rather than 90 %. It returns **the same three episodes and no others**, so the three are not an artefact of the threshold. By construction the criterion cannot see single-core losses or losses that run into a reconfiguration day; the OS counts those above.

Ruling out a collector fault. The competing explanation is that the collector dropped exactly those two metric streams. We test it directly: for each episode we ask whether the *other* sensors on the same node, in particular on the same socket, kept reporting. They did. In all three episodes the two slice sensors read zero for the entire window while **all ten control metrics, including `pX_power` and `pX_vdd_temp` on the same socket, sustained a full 4,320 samples per day** (Fig. 13). A collection fault that silences precisely the two cores of one slice for several consecutive days, while leaving every other sensor on the same socket at full cadence, is not a plausible account.

And they sit across reboots, as the mechanism requires. A GARD record is applied by hostboot at IPL, so a deconfiguration, and its later clearing, should each land on a boot. On the independent `boottime` signal of Sec. IV-H, node 946's episode of 2020-08-31–09-01 falls between boots on 08-30 and 09-03, node 411's of 2021-11-13–15 between boots on 11-12 and 11-15/16, and all nine OS-confirmed deconfigurations begin on a boot day. Node 347's March episode predates `boottime` collection (2020-05-05): absent coverage, not an absent reboot.

We therefore attribute these episodes to firmware deconfiguration with the signature of a GARD record applied at boot; the record itself was not read. We are not aware of an earlier report of such episodes recovered from fleet telemetry.

TABLE IV
CORE-GUARD EPISODES MEETING THE STRICT CRITERION.

Socket	Dates	Days	Slice lost	Afterwards
347 p1	2020-03-13–16	4	4 (cores 8,9)	same map
411 p1	2021-11-13–15	3	6 (cores 12,13)	new map
946 p0	2020-08-31–09-01	2	8 (cores 16,17)	new map

Baselines are the adjacent observed day. *Afterwards*: the socket's first clean day (347: 2020-03-18; 411: 2021-11-17; 946: 2020-09-08).

Changes never occur during operation: Of **235** transitions across the full record, **zero** occur between consecutive reporting days: every one coincides with a gap in reporting (median 2 d), against a background in which only 4.2% of consecutive reporting days are more than one day apart.

A reporting gap is not a reboot, so we measure the reboot: That argument infers a reboot from a *gap*, and collectors stop for many reasons. The inference is unnecessary: the Ganglia plugin publishes `boottime`, the epoch at which a node last booted, so a reboot is directly observable as a change in that value. Extracting it for all 31 months gives 668,524 node-days across 990 nodes, of which 1.74 % carry a reboot.

Of the **132 steady-state transitions with boot coverage, 126 (95.5 %) coincide with an observed reboot**, against **47.6 %** expected once control intervals are matched on length (49.0 ± 4.6 of 103 node-level transitions; the raw per-node-day rate is 1.74 %, but a transition sits across a multi-day reporting gap, so it is not the comparator). The length-matched comparison is Table V, and the result survives the dependence between intervals: permuting transition labels only *within* node $\times$ interval-length strata, with a node's two sockets as one unit, gives 99 of 103 node-level transitions with a reboot against 49.0 ± 4.6 expected (exact conditional $p = 6 \times 10^{-31}$), a length-standardised excess of $+63$ points (node-cluster bootstrap 95 % CI $[+57, +68]$).

The claim is therefore no longer that changes coincide with missing data, but that they coincide with *boots* [OI-14], measured by a collector that never touches the IPMI sensor family. The remaining six are four node-level events whose Ganglia stream stopped part-way through the interval and resumed only on 2021-05-19; the boot epoch then reported lies *inside* all six intervals. Dating reboots by epoch rather than by the day first seen, **133 of the 136 steady-state transitions contain a boot and none is observed without one**; three lack any bracketing `boottime` observation.

TABLE V
MAP TRANSITIONS AGAINST AN INDEPENDENT BOOT SIGNAL, MATCHED ON INTERVAL LENGTH.

Interval	Transitions with reboot	Controls with reboot
2 d	93.0 % ($n = 71$)	35.8 % ($n = 16{,}055$)
3–4 d	100 % ($n = 20$)	16.7 % ($n = 21{,}953$)
5–8 d	100 % ($n = 22$)	12.9 % ($n = 1{,}864$)
$\geq$9 d	94.7 % ($n = 19$)	66.3 % ($n = 5{,}912$)

Reboots flagged by day. No transition has a 1-day interval, which is the same fact as "zero occur between consecutive reporting days". Dated by boot epoch, every transition row is 100 %.

The only two apparent counterexamples confirm the rule once examined at full timestamp resolution. Nodes 315 (p1) and 342 (p0) appear to change within a single day, producing 20- and 22-core socket-days. In fact both cut over on 2021-03-02 across a *sub-day reporting gap*: on node 315 the outgoing cores $\{4, 5, 20, 21\}$ report until 08:42 and the incoming cores $\{8, 9, 10, 11\}$ resume at 09:10, a 28-minute outage, with exactly 16 cores before and 16 after; the 20/22-core counts are unions across the cutover. Node 342 cuts over in the same window, and `boottime` records a reboot of both nodes that morning (09:10 and 09:13).

Staticity as a prediction, not a description: The figures above describe the record in-sample. We also test them out-of-sample: derive each socket's map from the *first* half of the record (2020-03 to 2021-06) and use it to predict every clean socket-day in the *second* (2021-07 to 2022-09). Over **839,067 held-out socket-days the prediction is correct 96.77 % of the time** (95 % CI $[96.73, 96.81]$), and the errors are not diffuse: they fall on **89 of 1,962 sockets** (4.5 %), which are the sockets that genuinely reconfigure between the two halves. A map measured once therefore forecasts a socket more than a year later.

Changes concentrate at commissioning: The 235 transitions involve 132 nodes on 81 distinct dates, but 2020-03-20 alone accounts for 63 and the top three dates for 41 % (Fig. 14), an early, synchronised regime that the next paragraph locates and excludes.

Commissioning period, identified and excluded: Two independent signals locate the end of commissioning without assuming a date. Telemetry coverage, the fraction of socket-days that are clean, is 0.82, 0.00 and 0.40 in 2020-03, -04 and -05, then 0.72–1.00 in every later month but one (2021-03, 0.50). The transition rate falls from 91 in 2020-03 to a median of 4 per month thereafter, and the operator announced Marconi100's postponed start of production for 2020-05-04 [27]. We therefore treat 2020-03 to 2020-05 as commissioning and re-run every result on 2020-06 onward:

- **The harvest maps are unchanged**: for all 1,960 sockets with a map in both windows, the recovered map is *byte-identical* (100.0 %). Marginals agree to 0.1 percentage points and the pattern count is 443 either way.
- **The lot boundary is unchanged**: Lot A 88.2 %, Lot B 39.2 %, Welch $t = 26.6$.
- **Zero transitions during operation still holds**: of the 136 steady-state transitions, *none* occurs between consecutive reporting days.
- **The fleet-wide clustering disappears with the commissioning window**: the largest single-date cluster in steady state is 11 events (8 % of steady-state transitions), against 63 (27 %) on 2020-03-20.

Every structural and longitudinal claim in this paper therefore rests on the post-commissioning period, and none of them depends on the anomalous early months.

Some apparent changes are socket relabelling: Socket relabelling (RELABEL, Sec. IV-I) accounts for **19%** of the 105 steady-state node-level map-change events (20; 95 % CI $[12, 28]$) and 20% of the 230 socket-level transitions across the whole record where both sockets are observable.

Caveat: An apparent change can arise from (a) genuine hardware reconfiguration, (b) socket relabelling, or (c) a change in the BMC sensor-ID mapping, e.g. after a firmware update. Telemetry separates these only in part: a fleet-wide remap would impose one permutation of slice indices on every socket it touches, and neither the March 2020 cluster nor any steady-state date with four or more changes admits one (Sec. IV-I). The March cluster also lies in the commissioning period excluded above, and every result is reported on the steady-state period. The ambiguity is therefore bounded rather than load-bearing.

I. A taxonomy of map changes

A harvest map is carried by the die in a socket, so a change in the map is informative about what physically happened. A map is a signature, not a serial number, so we name each class of change by the event it is *consistent with* (Fig. 1, bottom). If p0 loses exactly the slices p1 gains and vice versa, the same two dies are present and only the tags moved (RELABEL). If exactly one socket's map changes while the other is observed and unchanged across the same interval, the socket received another processor (CPU-SWAP); if both change and it is not a mirror, both did (NODE-SWAP). A count dropping below 16 at a boot by one slice or one core is a deconfiguration with no die movement (GUARD, Sec. IV-H).

TABLE VI
MAP-CHANGE EVENTS, CLASSIFIED AT NODE LEVEL.

Kind	Commiss.	Steady	% of steady
RELABEL	3	20	19.0
CPU-SWAP	11	68	64.8
CPU-SWAP?	2	6	5.7
NODE-SWAP	40	11	10.5
Total	56	105	100

Node-level events (the 105 steady-state events are 136 socket transitions), each named by the event it is consistent with. RELABEL: tags swapped. CPU-SWAP: one socket received another processor; ?: partner unobserved. NODE-SWAP: both did.

The split (Table VI) validates the taxonomy on its own terms. During commissioning NODE-SWAP dominates (40 of 56, 71 %), since whole machines are being installed and

reconfigured. In steady state the picture inverts: CPU-SWAP (with CPU-SWAP?) accounts for 70 % of the 105 steady-state node-level events and NODE-SWAP for 11 %, which is what a maintenance regime that services individual processors looks like. Steady state contains 105 events on 87 distinct nodes over 28 months, about 3.8 per month.

Two consequences are worth stating. First, **19 % of steady-state map changes move no die**; a study that read every change as a replacement would overcount by that much. Second, counting one die per CPU-SWAP and two per NODE-SWAP gives **96 die changes** over 4,561 socket-years of calendar exposure (1,960 sockets over the 850-day post-commissioning window; a change is recovered by comparing maps *across* a reporting gap, so an unobserved day is still a day at risk): sockets **change die at 2.1 % per year**, 95 % CI $[1.6, 2.7]$ by rack-cluster bootstrap. The exact Poisson $[1.7, 2.6]$ is too narrow, since a NODE-SWAP moves two dies and changes cluster by rack (37 of 49 racks, $z = +2.55$, $p = 0.016$); observed clean socket-days as denominator give 2.4 %. Not every die change is a new processor. Because a map travels with its die, we *trace* a die when a socket's new map is exactly the old map of another socket whose map changed within seven days, one maintenance window. **50 of the 96** are traced (4.4 expected by chance; 42 of them on the same day): 43 come from another node, 36 of them in 18 two-way exchanges, and 7 from the partner socket. The other 46, new processors at an inferred **1.0 % per socket-year** (rack-cluster CI $[0.7, 1.4]$; 1.2 % if only same-day moves are traced), include all nine changes that follow an OS-confirmed deconfiguration. Other misreadings are small: a replacement die with the old map escapes with probability 2.1 % (Lot A) or 0.26 % (Lot B), and the data fit neither a firmware remap nor in-place reconfiguration. Only service records could measure each class's precision.

J. A clustering parameter, fitted per lot

Section IV-D establishes clustering against a null; it does not *quantify* it, and the yield models we invoke [1], [24] are parameterised by a clustering strength. Those models describe defect *counts* per die, which we cannot observe: the SKU fixes the count at exactly four harvested slices. What is observable is the conditional spatial arrangement given four, so we fit the natural conditional analogue, a pairwise-interaction model on the twelve slice positions: for a harvested set $S \subset \{0, \dots, 11\}$ with $|S| = 4$,

$$P(S) \;\propto\; \exp\Big(\textstyle\sum_{k\in S}\beta_k \;+\; \gamma\,\mathrm{adj}(S)\Big),$$

where $\mathrm{adj}(S) = |\{k : k \in S,\ k{+}1 \in S\}|$ counts index-adjacent pairs. The β_k absorb per-slice propensity (the slice-0 asymmetry) and γ measures adjacency *beyond* what those propensities explain; since $|S| = 4$ for every S, a constant added to all β_k cancels, so we impose $\sum_k \beta_k = 0$. The normalising constant is an exact sum over all $\binom{12}{4} = 495$ patterns, so the log-likelihood is exact and strictly concave, and Newton's method converges to a gradient norm below 10^{-8}.

TABLE VII
FITTED CLUSTERING PARAMETER BY LOT.

Population	n	γ	LR χ^2_1 vs γ=0
Lot A (racks 0–21)	880	$+0.743 \pm 0.047$	174.2
Lot B (racks $\geq$22)	1,082	$+0.434 \pm 0.058$	106.4
Whole fleet	1,962	$+0.647 \pm 0.047$	394.5

Uncertainties are rack-clustered bootstrap standard errors (2,000 resamples of the racks, within lot for the lot rows), not i.i.d. values, because sockets are not independent; see the text. Lot B includes node 980's two sockets (Table III); without them $\gamma_B = +0.430$ and the lot difference keeps $z = +4.02$. The one-γ model is misspecified (Sec. IV-J); γ summarises residual adjacency, not the full spatial structure.

Three things follow (Table VII). First, **residual adjacency is significant in both lots** (χ^2_1 of 174 and 106, $p \ll 0.001$; rack-clustered Wald z of $+15.9$ and $+7.5$): mean adjacent-pair counts are 1.463 and 1.218 observed against 1.203 and 0.993 under the no-clustering model. Second, **the lots differ significantly in clustering strength**, not only in their marginals. Here the uncertainty needs care: an i.i.d. likelihood over sockets gives $z = +4.47$, but sockets are *not* independent: each node contributes two, and sockets in a rack may share more than their lot (Sec. IV-G finds at most a marginal excess), so resampling racks is the conservative choice. Resampling *racks* rather than sockets changes the standard errors by $0.85\times$ to $1.47\times$ and puts the difference at $\gamma_A - \gamma_B = +0.309$, $z = +4.16$, 95 % CI $[+0.168, +0.455]$; a delete-one-rack jackknife ($z = +4.04$) and a rack-clustered sandwich ($z = +4.06$) agree. The claim survives every uncertainty model we tried; we report the clustered figure. Third, and most usefully, the two effects *separate*: Lot A's $\beta_0 = +2.51$ dwarfs every other propensity while Lot B's β vector is nearly flat (range $+0.34$ to -0.13), yet both carry a substantial positive γ. The slice-0 asymmetry and the residual adjacency are therefore statistically separable summaries of the maps rather than two views of one.

The model is useful and incomplete, and we show both: A likelihood-ratio test tells us $\gamma \neq 0$; it does not tell us the model is adequate. We therefore draw 10,000 replicate data sets from the fitted model, refit each (a parametric bootstrap, not a posterior-predictive check), and compare them to the data on quantities it was *not* fitted to. The model is fitted to the per-slice counts and the adjacency total, so the full pattern distribution, the number of distinct patterns, and the largest gap between harvested slices are all out-of-statistic checks.

It passes some and fails others. The count of distinct patterns is reproduced (443 observed against 436.4 ± 6.3 replicated, $p = 0.16$). But the pattern-frequency discrepancy is not: 955.6 observed against 481.5 ± 31.1 replicated (none as large, $p = 1/(B+1) = 10^{-4}$), and the observed maps are systematically *more compact* than the model predicts (mean maximum gap 4.469 against 4.685 ± 0.044, $p = 10^{-4}$). The residual co-harvest matrix (Fig. 10) shows exactly where it fails, and the failure is structured: because γ is fitted to the *total* adjacency, the model reproduces that total by construction but distributes it wrongly across the die, over-predicting co-

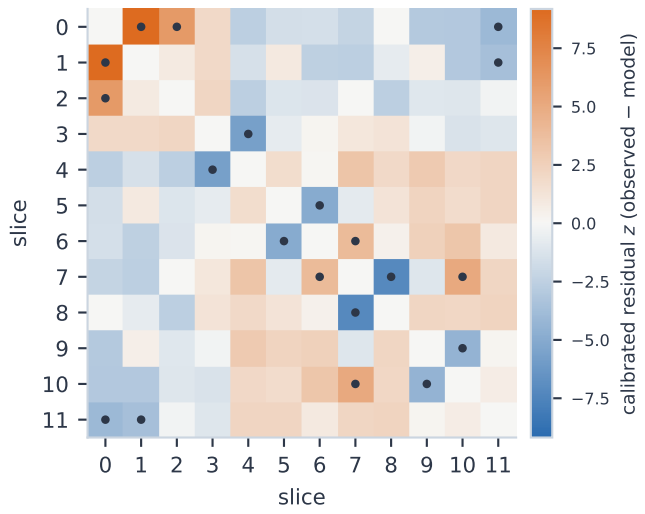


Fig. 10. Model-replicate residuals of the pairwise-interaction model: observed minus model-expected co-harvest counts, each standardised by its distribution over 10,000 refitted parametric replicates (simulation-calibrated z; dots mark cells outside the simultaneous 95 % band). The model is fitted to the *total* adjacency, so it matches that total but misplaces it, over-predicting at $(7,8)$, $(3,4)$, $(5,6)$ and under-predicting at $(0,1)$, $(0,2)$, $(7,10)$. The residual adjacency is real but not position-independent.

harvest at slices $(7,8)$, $(3,4)$ and $(5,6)$ (simulation-calibrated $z=-7.2$, -5.7, -5.0) and under-predicting it at $(0,1)$, $(0,2)$ and $(7,10)$ ($+9.2$, $+6.1$, $+5.0$); 10 of 66 cells lie outside the simultaneous 95 % band, and the $(7,8)$ miss recurs in each lot fitted alone. **A single, position-independent adjacency term does not capture the spatial structure**, and the miss is regular: adjacency within a quad (slices $2m, 2m{+}1$) carries $\gamma = +0.92 \pm 0.04$, across quads only $+0.28 \pm 0.06$ (difference $z = +13.9$, racks resampled), and even that split leaves misfit. The term still captures a large part of it (the same discrepancy against the no-clustering model is 1631.9), so γ summarises a real effect rather than an artefact. We report the misspecification because a reader otherwise has no way to tell a fitted parameter from an adequate model.

K. Sensitivity to the analysis choices

Every number above depends on three discretionary choices: where to set the clean-day threshold, how to summarise a socket's clean days into one map, and how many draws to run the null for. We varied all three (Table VIII).

The clean-day threshold is inert. Moving it from 0.50 to 0.99 discards 18 % of socket-days and changes nothing that matters: the same 1,962 sockets (1,960 at 0.99), 442–443 patterns, an identical 61.2 % slice-0 rate and an identical lot contrast. This is a stronger statement than "we chose 0.90".

The map rule matters slightly, and in the expected direction. Taking the first clean day instead of the mode changes 163 of 1,962 maps (8.3 %) and taking the last changes 49 (2.5 %); these are exactly the sockets that undergo a transition, and the asymmetry (more change under "first" than "last") reflects that commissioning-era configurations are the ones later replaced. The lot contrast ranges only over $t = 25.9$–28.4. The *strict* rule, which keeps only sockets whose every clean day is identical, retains 1,756 sockets, precisely the 89.5 % that never change, and still reproduces the boundary at $t = 28.0$.

TABLE VIII
SENSITIVITY TO THE ANALYSIS CHOICES.

Choice	Sockets	Patt.	$P(s_0)$	t
clean-day threshold				
0.50	1,962	442	61.2 %	26.6
0.80	1,962	442	61.2 %	26.6
0.90	**1,962**	**443**	**61.2 %**	**26.6**
0.95	1,962	443	61.2 %	26.6
0.99	1,960	443	61.2 %	26.6
map rule, threshold fixed at 0.90				
mode	**1,962**	**443**	**61.2 %**	**26.6**
first clean day	1,962	442	61.6 %	28.4
last clean day	1,962	443	61.1 %	25.9
strict	1,756	436	60.8 %	28.0

Settings used in the paper are in bold. t is the Welch statistic for the lot contrast. Retained clean socket-days fall from 1,607,604 at threshold 0.50 to 1,324,982 at 0.99; *strict* keeps only sockets whose every clean day is identical. At 0.50 and 0.80 one pattern drops out: node 806 p0, its only carrier, holds its two main maps on 358 and 357 of 781 clean days, so its mode flips with the threshold.

The null is converged. Chains from the observed matrix and from a maximally structured one with the same margins agree within 30 global trades ($\hat{R} = 1.000$), and 10,000 draws of the sequential sampler give $z = -20.5$ against -20.7. The 200-draw null of earlier drafts used a trade that ran only between sockets sharing no harvested slice (10 % of attempts) and drew its first sample before the chain had forgotten its start; its null moments nonetheless agree (4.582 ± 0.017).

The thermal alignment grid was the one remaining untested choice; it has since been varied over 20 s–15 min and is reported in Sec. IV-E. It is the only choice we found that materially moves an effect size, which is why the affected number is now quoted at native resolution.

L. The map and socket power: bounded rather than dismissed

Recovering a hidden hardware property is only useful if it changes something. We therefore ask whether the harvest map predicts socket behaviour, using a **within-node paired contrast**: the two sockets of a node share chassis, airflow, inlet air and power supply, so the difference $p_0 - p_1$ removes the node-level confounds that make cross-node power comparisons unreliable. We summarise each socket's map by the mean index of its active slices, the number of adjacent active pairs, and the count of active slices in the low half, and regress the socket-to-socket difference in mean CPU power on the difference in each summary, one summary per model (Fig. 15). Each node-day carries the map in force that day, from the run of identical clean days containing it, rather than the socket's most common map over the whole record.

Pairing does not remove a workload the scheduler places unevenly between the sockets, and the node-level covariates we adjust for (utilisation, load average, inlet air, month) cannot measure it. The map, however, is fixed at boot and

invisible to the scheduler, which allocates 16 interchangeable cores per socket, so uneven placement biases the slope only through its correlation with the map difference. Where the job table records each job's cores (June–October 2020), placement explains 40 % of the variance of $p_0 - p_1$ but correlates with the map difference at only $|r| \leq 0.021$; adjusting for it moves each estimate about 1 W away from zero, and no interval extends beyond 5.0 W. Later job records carry no core layout, so the full record cannot be adjusted the same way. Idle node-days, with almost no workload to place, give a second check.

Failing to reject is not the same as showing negligible: An earlier version of this analysis used six months of power data, reported $r = +0.039$ (permutation $p = 0.003$) explaining 0.16 % of variance, and concluded that harvest maps are "operationally invisible". That conclusion does not survive two corrections, and we state the revision plainly because it changes a headline claim.

The first correction is the instrument. "No measurable effect" cannot be established by a small r; the question is whether the effect is small *enough*, which requires an equivalence test against a margin fixed in advance. We adopt ± 5 W per socket, 7.2 % of the 69.0 W mean draw and well inside the power spread of up to 23 % measured between nominally identical processors [28]. Any margin is a judgement, so Table IX also gives the smallest margin each interval clears, which settles the verdict at ± 2 and ± 10 W as well. The second correction is the uncertainty: node-days within a rack are not independent, and the previous interval treated them as if they were.

Re-run over the **28 steady-state months (636,026 paired node-days on 980 nodes)** with contemporaneous maps and rack-clustered bootstrap intervals, the result is (Table IX):

TABLE IX
EQUIVALENCE TESTS FOR THE MAP'S ASSOCIATION WITH $p_0 - p_1$ SOCKET POWER: COVARIATE-ADJUSTED, ONE SUMMARY PER MODEL, CONTEMPORANEOUS MAPS, RACK-CLUSTERED.

Summary	Slope (W/unit)	Full range (W) effect	90 % CI	Per SD (W)	Smallest margin
mean slice index	+0.40	+2.7	$[-0.2, +5.6]$	+0.46	5.6 W
low-half count	−0.19	−1.5	$[-4.7, +1.8]$	−0.25	4.7 W
adjacent pairs	−0.14	−1.0	$[-4.0, +2.1]$	−0.17	4.0 W

Effect = slope × the full observed range of the $p_0 - p_1$ difference in each summary (6.75, 8 and 7 units; SD 1.14, 1.31, 1.19). A 90 % interval inside $\pm m$ is the two-one-sided-tests criterion at $\alpha = 0.05$: each summary is equivalent at any margin above the last column.

Two of the three summaries are equivalent at ± 5 W, all three at ± 10 W. The mean active-slice index misses ± 5 W narrowly: between the most dissimilar maps observed its interval runs from -0.2 to $+5.6$ W, only just including zero. The previous version's $+7.9$ W $[+0.9, +16.6]$ came from fitting the three summaries jointly and scaling each partial slope by its marginal range, a contrast no real pair of maps realises (mean index and low-half count correlate at -0.89); over the map pairs that do occur, the same joint fit spans 2.2 W from the 5th to the 95th percentile pair. For scale, $p_0 - p_1$ has mean 6.8 W and SD 19.9 W. The mean reflects scheduler placement rather than socket position: at equal placement the slot effect is $+0.3$ W, $[-0.4, +1.1]$, and on idle node-days the mean is $+0.2$ W. Even idle, however, the two sockets of a node differ by 9.2 W (SD of node means), a static die, board or sensor offset.

Across the other designs (Fig. 15; also same-day maps only and days with full power coverage) every interval lies within ± 7.4 W, but not all within ± 5 W. The 848 nodes whose maps never change, which excludes every relabel and map-changing replacement, give mean index $[-0.1, +6.5]$ W, and on idle node-days adjacency gives -5.0 W, $[-7.2, -0.5]$, the one design in which an adjusted interval excludes zero; it weights nodes by idle time, and with one observation per node-epoch it is -2.5 W, $[-5.1, +0.1]$. The summary is therefore *bounded and small, but not provably negligible*: a few watts between the most dissimilar maps, below the spread between sockets, yet not inside ± 5 W for every summary in every design. The claim that the map is operationally invisible stays withdrawn. Note also that this does not weaken the provenance argument of Sec. IV-G: the lot signature travels with the dies, not with the rack they sit in.

M. The map does not usefully predict which dies get replaced

The map is set before a die enters service; replacement is what the field eventually concluded about it. If dies whose harvested slices sit in a tight cluster were the ones that failed, that would tie manufacturing variability to operational reliability far more directly than any power correlation. We test it.

The design is a discrete-time survival model in calendar months (June 2020 to September 2022; all but two dies enter in the first month, so this is also time since entry). A die is at risk from its first steady-state clean day until its event; every other die is followed to the end of the record, so censoring is purely administrative. The event, inferred from the map rather than read from a service record, is the first month in which the die's map leaves its node: a CPU-SWAP or NODE-SWAP in which the map is not re-found on the partner socket. Dies are followed by their map, so a RELABEL moves no die, and two dies leaving together are two events. The interval hazard $h_i(t) = P(T_i{=}t \mid T_i{\geq}t)$ is $1 - \exp\{-\exp(\alpha_t + x_i^\top \beta)\}$, the grouped proportional-hazards model [29], so e^β is a hazard ratio; α_t is one baseline effect per month, shrunk towards their mean, as monthly events range from 0 to 15. Covariates come from the die's *first* steady-state map: slice-0 status, the number of index-adjacent harvested pairs, the dispersion of the harvested indices, the rarity of the pattern ($-\log$ of its smoothed frequency) and the lot. This gives **85 events over 53,716 die-months**, a monthly hazard of 0.158 %; 44 of these dies were never seen again and 41 reappeared on another node, traced as in Sec. IV-I. (The 96 of Sec. IV-I also counts 7 dies re-found on the partner socket and 4 later departures of dies already counted or installed during the record.)

No feature is a significant predictor. No hazard ratio's rack-clustered bootstrap interval excludes 1: 1.25 $[0.51, 3.26]$

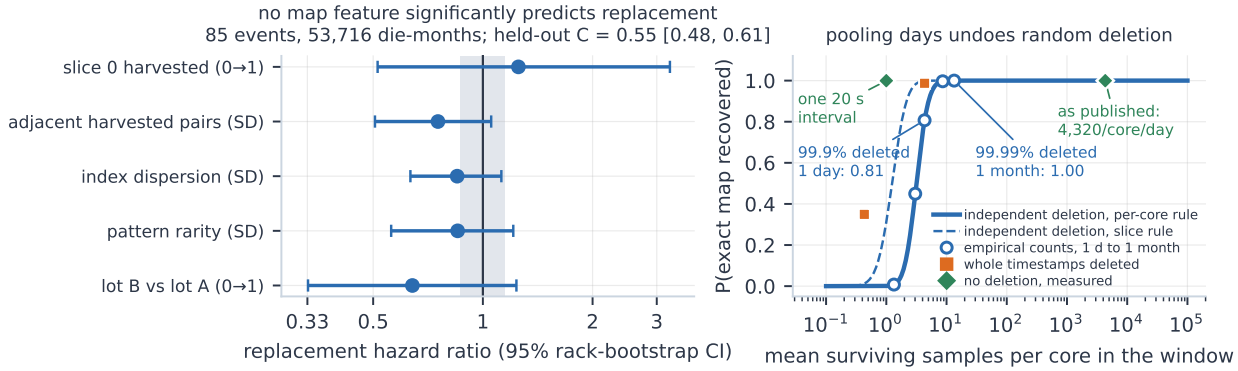


Fig. 11. Two bounded nulls. **Left:** no feature of the baseline harvest map is a significant predictor of subsequent replacement; hazard ratios (0→1 for the binary features, per standard deviation otherwise) with rack-clustered bootstrap intervals, shading marking a $\pm 15\,\%$ negligible-effect band. Held-out risk-set concordance is 0.55 $[0.48, 0.61]$ against a rack-permutation null of 0.50 ± 0.04. **Right:** the cost of the disclosure, against the mean number of samples per core surviving into the window. Curves: independent deletion, with a surviving sample needed on every active core (solid) or on every active slice (dashed); circles: 99.9–99.999 % deletion of the empirical counts over one day to one month (Monte Carlo agrees); squares: whole timestamps deleted; diamonds: no deletion, one 20 s interval and one day (squares and diamonds on raw timestamps). Pooling days moves every deletion rate to the right, and the dataset ships 4,320 samples per core per day; reducing resolution is not a mitigation.

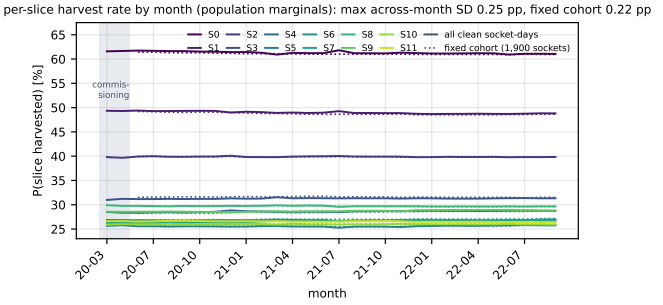


Fig. 12. Per-slice harvest rate by month: all clean socket-days (solid; 2020-04 has none) and a fixed cohort of 1,900 sockets seen every post-commissioning month (dotted). Largest across-month standard deviation: 0.25 percentage points (cohort 0.22). This is population stability; per-socket staticity is in Sec. IV-H.



Fig. 13. The three strict-criterion core-guard episodes against every other collector that can see them. The grid is BMC per-core presence (pale = not configured, blue = configured and reporting, light blue if partly sampled, orange and hatched = configured but silent, purple and lavender = the new and the replaced map); the strips below are the operating system's own logical-CPU count and the independent boot signal. In every case exactly one slice goes silent while all ten control sensors on the same node stay at full cadence; where coverage exists, the OS drops to $128 - 8 = 120$ over a window that brackets the BMC one, and the episode sits across observed reboots, as a GARD record applied by hostboot at boot would; the record itself was not read. Node 347 predates Ganglia collection, so its strips are marked as absent coverage rather than left blank.

for slice-0 status and 0.64 $[0.33, 1.24]$ for lot B (both 0→1), and per standard deviation 0.75 $[0.51, 1.05]$ for adjacent harvested pairs, 0.85 $[0.63, 1.12]$ for dispersion and 0.85 $[0.56, 1.21]$ for rarity, point estimates unchanged in the second decimal under a constant or spline baseline. Under seven-fold cross-validation grouped by rack and repeated over ten partitions, with rarity and scaling learned in the training folds, the risk-set concordance (each month's events against the dies of the same held-out fold still at risk that month [30]) is **0.55** (95 % CI $[0.48, 0.61]$) against a rack-permutation null of 0.50 ± 0.04 ($p = 0.12$; Fig. 11, left). Adding the map does not improve the held-out survival log-likelihood over the monthly baseline alone (-2.1 nats, $p = 0.14$), nor the integrated Brier score.

We report this as a bounded null rather than an absence of evidence [OI-16], and the bounds are asymmetric: the intervals exclude a hazard increase above 21 % per standard deviation of any continuous feature, but not a protective effect down to about 0.5, and for the binary slice-0 and lot contrasts they exclude only effects beyond about threefold. What the data do exclude is useful prediction of these events from these features: even the upper confidence bound on concordance, 0.61, lies below the 0.7 conventionally required for acceptable discrimination [31], and it stays below 0.7 for dies never

seen again (0.66), for CPU-SWAP-like events alone (0.59) and for the broadest endpoint, which also counts relabelling (0.64; its nominal rack-permutation $p = 0.015$ does not survive correction for six endpoints, `LIMITATIONS.md`); NODE-SWAP-like events alone (13) are too few to bound. Two readings are consistent with it: whatever set the disabled positions is unrelated to what kills a part in the field, or replacement is dominated by causes unrelated to the die. Either way, an operator should not use the harvest map as a reliability covariate, which, with Sec. IV-L, completes a consistent picture of a property that is highly informative about *provenance* and at most weakly informative about *behaviour*.

N. How much telemetry does the disclosure cost, and what closes it?

We have shown the maps are recoverable; an operator publishing such a dataset needs to know how cheaply, and which mitigations would work. Both follow from one asymmetry: a harvested core can never emit a sample, so the channel has *no false positives*. Recovery fails only if an active core stays silent for the whole window, so the question is not how long one must watch but how many samples per core survive collection.

Under healthy collection that question is almost vacuous. The median core reports **4,320 of 4,320** expected samples per day, so it appears in essentially every 20 s slot. Measured on the raw timestamps of 2022-08, a **single sampling interval recovers the exact map in 99.95 % of socket-intervals**, and in all but 5×10^{-7} of those in which the socket reports at all. From the daily counts, again without any model, exact recovery holds for 99.9 % of sockets at *every* window from one day to the full month. Modelling the degraded regime, where the number of samples surviving from each core is Poisson with mean λ, independently across cores, exact recovery is $(1-e^{-\lambda})^{16}$: negligible at $\lambda \leq 1$, 0.44 at $\lambda = 3$, 0.90 at $\lambda = 5$ and 0.99 at $\lambda = 7$. The knee is around five to seven *surviving* samples per core, and half that for an attacker who uses slice granularity: one surviving sample per active slice suffices (the *slice rule*).

That immediately rules out the obvious defences. Publishing at a coarser cadence does not work: keeping one 20 s slot in every six hours still recovers every map from a single day, and one slot per day 99.95 %, because a single sample names its core just as well as 4,320 do. Random deletion does not work either until it becomes destructive, because maps are static and an attacker pools days. Deleting each sample independently with probability 99.9 % leaves one-day exact recovery at 0.81 (0.9985 under the slice rule); at 99.99 % a month still recovers 0.99997 (Fig. 11, right), and holding one-month recovery below one half takes 99.998 % deletion, or 99.9991 % against the slice rule. **Reducing resolution is not a mitigation, because the channel is carried by which sensors *exist*, not by what they report.** Only defences that remove the per-core position work: publishing socket-level aggregates, or renumbering survivors compactly to $0 \ldots 15$, which is precisely what this BMC does not do. Per-node random renumbering destroys the map while still leaking the count, but must be redrawn for every release: a fixed permutation turns the reporting set into a fingerprint unique for 99.7 % of sockets, against 4.9 % for the maps themselves.

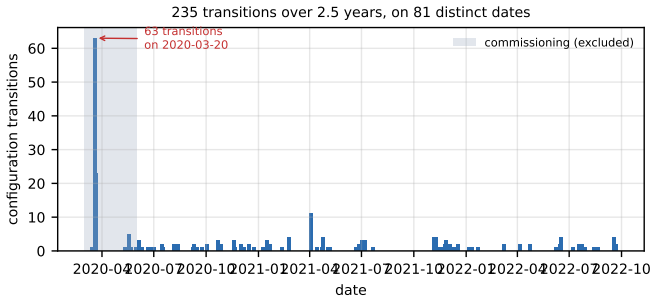


Fig. 14. Configuration transitions over the full record. 2020-03-20 alone accounts for 63, all inside the commissioning window (shaded) that Sec. IV-H excludes; the steady-state period shows no comparable cluster.

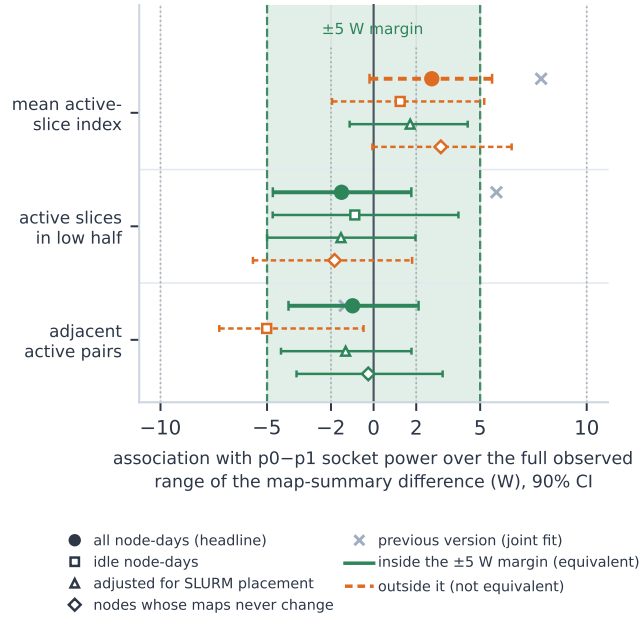


Fig. 15. Equivalence tests for the harvest map's association with $p_0 - p_1$ socket power. Points are the effect over each summary's full observed range with rack-clustered 90 % intervals, on one scale for four designs: all 636,026 paired node-days (filled), idle node-days, June–October 2020 adjusted for SLURM core placement, and the 848 nodes whose maps never change. Green intervals lie inside the ±5 W margin (band); dotted lines mark ±2 and ±10 W. Grey crosses: the previous version's joint-fit estimates.

V. Discussion

Why is slice 0 special?: The single most striking number is that slice 0 is harvested on 88.2% of Lot A sockets but only 39.2% of Lot B. Two explanations are live. Under *defect-driven* harvesting, disablement follows wherever killer defects land. Under *policy-driven* harvesting, slice 0 abuts a shared structure and is preferentially disabled for reasons of yield policy, binning recipe or physical design; an 88% rate on one specific slice is very hard to obtain from random defects.

Intel's Xeon Phi has four such favoured positions, read as a default set disabled on otherwise good dies [5]. A factory core-count override is not the explanation either: hostboot keeps the lowest-numbered core pairs [13], so, if the BMC index follows firmware numbering, an override would spare slice 0 and fall on slices 8–11, yet in both lots slice 0 is harvested most often and slices 8–11 are not conspicuously above average (Lot A 20–25 %, Lot B 31–35 % against 33.3 %).

The fit of Sec. IV-J shows that *two statistically distinct effects are present*, and estimates them separately. The propensity vector β carries the slice-0 asymmetry and differs enormously between lots ($\beta_0 = +2.51$ versus a flat Lot B); the interaction parameter γ carries residual spatial association and is significantly positive in *both* lots. The most economical reading is a change of binning recipe or die source between the two lots, superimposed on a spatial process present in both.

We are careful about what this does *not* establish. The decomposition is descriptive, not causal: a positive γ says harvested slices are spatially associated beyond what the per-slice propensities explain, and a correlated *defect* process and a binning *policy* that disables contiguous blocks both predict exactly that. The model cannot tell them apart, and neither can any amount of data from one machine. Nor is the model adequate on its own terms (Sec. IV-J): it under-predicts how compact the observed maps are, so even the split between "propensity" and "clustering" is a summary of the structure rather than a complete account of it. What survives is that the two lots differ in *both* parameters, which is a statement about the populations, not about the mechanism.

What the rack stands for: The rack-position objection is tested directly in Sec. IV-G; what rack order can still stand for is the parts' history up to installation, which is the provenance signal we claim and which telemetry cannot decompose further.

How identifying is a harvest map?: Since the map is a per-die property, it is fair to ask whether it identifies a socket, which would raise a different set of questions from the provenance ones we pursue. Across a whole fleet, one map does not. Over 1,962 sockets the pattern distribution carries $H = 8.08$ bits (plug-in estimate, log base 2), or 8.24 and 8.33 bits after Miller–Madow and Chao–Shen bias correction (rack-jackknife SE 0.12), an effective $2^H = 270$–321 distinct maps against 495 possible; only **4.9 % of sockets (97) have a map unique in the fleet**, and a typical map is shared by 4.4 sockets. Uniqueness grows quickly with side information, though: to 12.1 % with the lot, 78.4 % with the rack, and a node's two maps together are unique for 93.0 % of nodes. A single map is thus a *partial* identifier, strong evidence about which population a part came from; a node's pair of maps is a quasi-identifier, which matters to anyone publishing such telemetry under pseudonymised node names. It records what was disabled rather than an identity designed into the die, which is why this work sits apart from the fingerprinting literature [32] rather than extending it.

Operational uses: Fleet telemetry can detect latent hardware cohorts without vendor cooperation or an asset or procurement database, testing whether the installed parts are one population and, per Sec. IV-G, placing 96 % of held-out racks in the right lot. It can detect field deconfiguration events that would otherwise be visible only in service logs, and tell dies moved between nodes from new ones. And it *bounds* how much a hardware property matters: the map does not usefully predict replacement (Sec. IV-M), and its association with socket power is bounded to a few watts (Sec. IV-L). Bounding an effect is a weaker claim than dismissing one, and the right one to make here.

VI. Related work

Yield, salvage and segmentation: Yield models treat defect statistics directly: the negative-binomial family [1] and spatial wafer-map models [24] describe how many killer defects a die carries and how they cluster, and are fitted to fab data unavailable to system owners. Disabling a faulty core so that the die still ships is the baseline response in multicore design [2], [3], and vendors also disable working units to create cheaper products [4], [5]. We observe neither the defects nor the reasons, only which positions end up disabled, and only conditionally on the SKU's fixed count of four.

Variability in HPC: Manufacturing variability in HPC has been studied through its *performance* consequences: frequency and power spread under a power cap [28], turbo-boost variation across nominally identical processors [33], and accelerator-level variability in GPU-rich systems [34]; field-reliability studies track components by serial number [35]. That literature treats the die as a black box characterised by its behaviour, or relies on an asset inventory; we recover internal structure from telemetry alone.

Provenance and disabled-unit maps: Physical unclonable functions and inherent hardware identifiers exploit analog manufacturing variation to answer *which die is this*, for traceability and counterfeit detection [32]. That is a different question: we do not identify a die, we recover *what was disabled on it*, without designed-in circuitry or physical access. Closest to us is McCalpin: running a user-mode instruction on all 4,200 Xeon Phi processors of Stampede2, he recovered each die's disabled core pairs, separated a default disable set from defects, found that the defect rate differed between the machine's two deployments and counted processor replacements between two snapshots [5]; on its Xeons he read disabled tiles from a privileged register [8]. Smith read the enabled cores of one POWER9 machine from the same BMC sensors we use [10]. Our source is public out-of-band telemetry, with no access to the machines, followed daily for 31 months, and we locate the cohort boundary from the maps rather than from deployment records. Node-level anomaly and outlier detection in HPC [36], [37] likewise establishes *that* nodes differ without recovering the hardware structure responsible.

Telemetry side channels: Temperature sensors are established covert and side channels: their readings carry data between cores [38] and leak a kernel's address-space layout [39]. Those channels live in sensor *values*; the one here is carried by which sensors *exist*. The M100 ExaData release itself has

an active downstream literature spanning telemetry foundation models, job power data and operational-data ontologies [40], [41], [42]; none of it examines per-core sensor presence.

We are not aware of prior work recovering per-die harvest maps from published out-of-band telemetry, without access to the machines, or tracking them daily for years, and that is the gap this paper addresses. We state this as an awareness claim rather than a priority claim: it rests on a systematic but necessarily incomplete search.

VII. Threats to validity and open issues

Every item below carries the identifier it is tracked under in `LIMITATIONS.md` in the artefact repository, which records for each one what is known, what would settle it, and where to look; identifiers cited in the text but not itemised here (OI-14, OI-16) are recorded there too. Nothing here is load-bearing for a headline result unless the entry says so.

1) **Sensor presence $\neq$ core existence. Resolved externally for the count** (Sec. III-B): the operating system reports 128 logical CPUs on 99.9853 % of 847 M samples, the scheduler's configured totals agree, and all nine OS departures from 128 are matched by a BMC loss of that many cores, all independently of the IPMI sensor family. Neither collector sees which positions are active; those rest on IBM's core numbering (OI-1).
2) **[OI-1] Resolved by documentation, with one link from a non-vendor source.** IBM's chip diagram places the cores and labels the chiplets EC00–EC23 [15, Fig. 2]: the two slices of a quad are neighbours, index neighbours in different quads are not. That the per-core sensors carry those numbers is reported in [10] and is consistent with every structural result here, none of which would survive a compact renumbering of the survivors. Sec. IV-F states what thermal coupling does and does not add; the diagram is a chip diagram, not a to-scale die photograph, and a vendor statement or a second AC922 installation would close the link.
3) **[OI-2] Bounded.** A March 2020 BMC firmware update could have remapped sensor IDs, mimicking reconfiguration; 86 of the 235 transitions fall on 2020-03-20–21, though no single slice permutation carries their old maps onto the new ones. Those days lie in the commissioning window excluded in Sec. IV-H; maintenance logs would confirm the cause.
4) **[OI-4] Partly resolved.** Every reference has been checked against its publisher's DOI record or the primary document, and each correction is logged in `LIMITATIONS.md`. The **novelty check** stays open by nature: our claim is an awareness claim, resting on a systematic but necessarily incomplete search, and the closest prior work we found [5], [10] is discussed under Related work.
5) **[OI-3] Defect-driven versus policy-driven harvesting is not identified.** A correlated defect process and a binning policy that disables contiguous blocks both predict $\gamma > 0$, and the one-γ model is incomplete (Sec. IV-J); an earlier draft's claim that the fit showed "both" is withdrawn. A second installation, or vendor disclosure, would be needed to separate the mechanisms.
6) **[OI-5] Resolved.** Monthly coverage ranges from 18 % to 99.4 %, but coverage is uncorrelated with the harvest pattern and restricting to $\geq$90 % coverage leaves the lot contrast unchanged (Sec. IV-G).
7) **[OI-6, OI-7] both resolved.** The sub-16 socket-days were dominated by an extraction defect on our side, now gated; the nine fully-sampled even-count days that survive are the three strict guard episodes of Sec. IV-H, and the odd-count days are single-core deconfigurations or single-sensor dropouts. The two nodes appearing to change within a day are same-day reboots.
8) **[OI-8] Largely settled.** Of the relabelling events whose socket power difference exceeds 2 W on both sides, **12 of 15 flip sign** at the relabelling instant (95 % CI $[52, 96]$ %), as a collector-side tag swap predicts; per-socket longitudinal studies on this dataset should still treat relabelling as a hazard.
9) **[OI-12] The guard episodes are a small count**: three meet the strict slice criterion, and the nine slice or single-core deconfigurations in steady state are each matched by the OS count and begin on a boot; the three strict ones are also corroborated by every other sensor keeping full cadence. The GARD record itself is not in the telemetry.
10) **[OI-15] Pair-level counts depend on how the null treats the two lots.** The 16–18 significant pairs of earlier drafts (five seeds of a 200-draw null) are now 20 against 100,000 exchangeable draws. A null that keeps each lot's own marginals leaves 11, so the distant deficits in particular are partly lot mixture; the near excesses (eight pooled, seven per lot) are the robust part (Sec. IV-D).
11) **[OI-11] Common-mode artefact** in the thermal analysis, addressed by same-distance contrasts and by ordering tests it cannot bias. The per-socket matrices, with the support of every cell, are part of the artefact, and every thermal interval resamples the 49 racks.
12) **[OI-13] Analysis choices moved our numbers more often than the data did.** *Five* reported results changed materially during this work, none because of a new measurement. *Sample size:* a three-day sample overstated the within/cross-slice contrast more than threefold ($+0.51$ vs $+0.29$, against $+0.372$ vs $+0.312$ over the full month) and reversed which candidate layout scored best. *Preprocessing:* an alignment grid, adopted to work around an unrelated bug, inflated that contrast by $1.3\times$ at one minute and $2.2\times$ at 15 minutes (Sec. IV-E). *Bookkeeping:* an intermediate revision quoted a corrected figure from a different subsample than the one it named. *Uncertainty:* the claim that the map is operationally invisible rested on an interval that treated node-days within a rack as independent; corrected, the effect was not bounded below the margin, and the claim was withdrawn. *Specification:* fitting three collinear summaries of one map jointly, and scaling each partial slope by its

marginal range, nearly tripled the reported power effect (+7.9 against +2.7 W; Sec. IV-L). **Four of the five were introduced while fixing or extending something else, and the first four all ran in the direction we would have preferred.** The defences that worked were mechanical: one stated pipeline for every number, varying each discretionary choice (Sec. IV-K), clustering uncertainty where the data are dependent, and reporting the most conservative variant. Anyone mining a telemetry archive of this kind should be at least as suspicious of their corrections as of their original analysis.

13) **[OI-10] Single system.** Findings may reflect CINECA's parts rather than POWER9 generally, although the same non-renumbered per-core sensors appear on another POWER9 platform [10]; a second AC922 installation is the most valuable next test.

VIII. Conclusion

Out-of-band telemetry that supercomputer operators already collect inadvertently publishes a vendor-undisclosed attribute of every processor: which of its units are disabled. For 1,962 POWER9 sockets it shows that every disabled core in a map lies in a fully disabled slice, with no exception on 1.52 M clean socket-days; that the maps are per-die rather than per-SKU, with 443 of 495 possible patterns realised; that harvested slices cluster above all within the quads, slice pairs adjacent in IBM's chip diagram, where thermal coupling is also strongest; and that a sharp changepoint at rack 22 separates two hardware populations, most plausibly procurement lots, placing 96 % of held-out racks in the right lot. Over 2.5 years the per-slice rates are stable to within 0.25 percentage points per month, maps change rarely (0.21 % of cohort socket-months) and at reboots (133 of 136 steady-state changes contain a boot, none is seen without one), and the nine steady-state core-guard episodes each begin at a boot and match the OS's own CPU count.

What remains interpretation, pending vendor, service or delivery records, is why each position was disabled, that the two populations are procurement lots, and that map changes are the events they resemble. Two things deserve emphasis. Thermal coupling confirms the processor's quads from telemetry alone, but beyond them it follows how the surviving cores are numbered, so the die layout rests on IBM's documentation, not on telemetry. And the harvest map's operational footprint is now *bounded* rather than dismissed: over 636,026 paired node-days its association with socket power is a few watts between the most dissimilar maps, and it does not usefully predict replacement. These are bounded nulls we consider as useful as the positive ones, since they tell operators what they can safely ignore and what they cannot yet. The broader point is that which sensors exist, not what they read, can expose stable component-level configuration and fleet cohort structure that aggregate telemetry would conceal, and that those who publish such datasets should treat sensor identity as data.

Ethics and disclosure

We used only the public M100 ExaData release (CC BY 4.0) [9], [20], passively: no running system was accessed, no access control bypassed and no personal data used, and Marconi100 left production in June 2023 [43]. The maps disclose a vendor-undisclosed configuration attribute and the fleet's cohort structure, not credentials or access. A node's pair of maps is nearly unique in the fleet, so it could link records of the same machine across datasets; the released tables carry only the dataset's own node identifiers. Sec. IV-N states which publication practices remove the channel.

Artefact availability

All analysis code, the derived per-(node, socket, core, day) tables, and the recovered harvest maps are available at the artefact repository accompanying this paper, together with `LIMITATIONS.md`, the register the identifiers in Sec. VII refer to. Every result is reproducible from the public M100 ExaData Zenodo releases [20]. The derived tables (15 MB) suffice for all structural and longitudinal results without re-downloading the 49.9 TB source. Two sections need one further pass over the archives, and the script that performs it is included: the thermal analysis reads the raw per-core series (its per-socket matrices are included, so every thermal number re-derives without it), and the reboot and power sections read a per-(node, day) summary of `boottime`, `cpu_speed`, `cpu_user`, `load_one` and `ambient`, and the power section also the per-socket core placement recorded in the SLURM job table.

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